

Switching

CCNA 200-301 V1.1

MODULE 3

Topics

3.1 Ethernet Basics

3.2 Switching Overview

3.3 Spanning-Tree Protocol (STP)

3.4 Rapid PVST+

3.5 VLANs

3.6 VLAN Trunking

3.7 VLAN Trunking Protocol (VTP)

3.8 VLAN Routing

3.9 Switchport Configuration

3.10 EtherChannel

3.11 Review



3.1 Ethernet Basics

CCNA 200-301 v1.1

Module 3

What is Ethernet?

Ethernet Frame Structure

Network Contention

Collision Domain

Broadcast Domain

What is Ethernet?

Most common Layer 2 wired LAN protocol

Uses source and destination MAC addresses

Can be carried on twisted pair, coax, or fiber optic cable

Uses Carrier Sense Multiple Access/Collision Detect (CSMA/CD) for contention management

Minimum length of 64 bytes

Default maximum transmission unit size (MTU) of 1500 bytes

Can be modified up to 9000 bytes (jumbo frames) for special purpose

- Better performance for an iSCSI SAN
- Requires switches and NICs that support jumbo frames



Ethernet Frame Structure

Uses physical (MAC) addresses for communication

- Source and destination for that segment

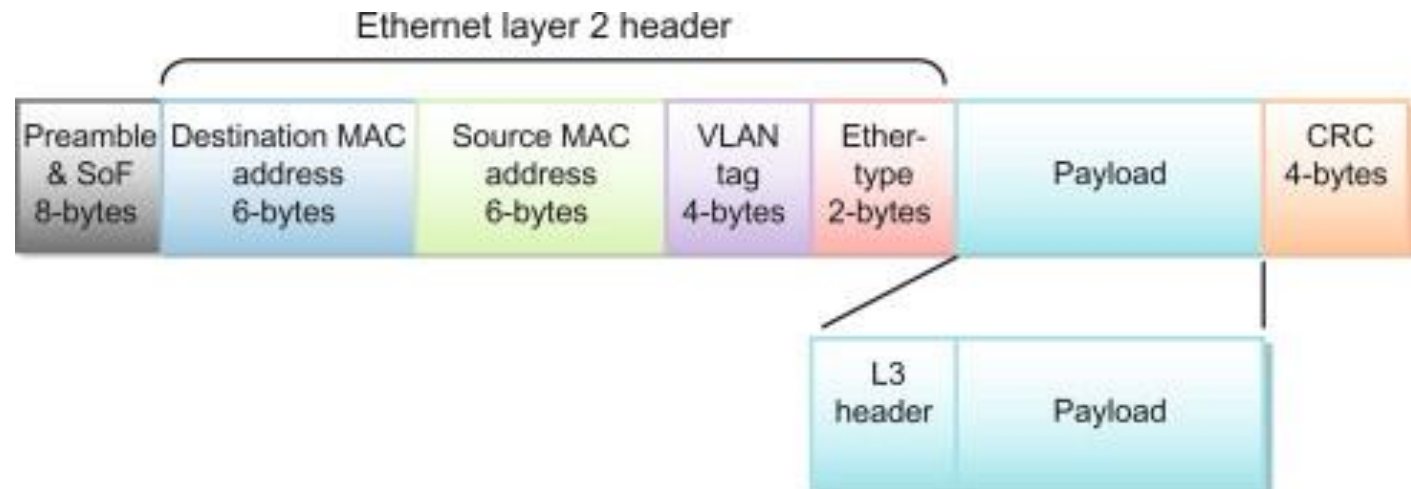
Ethernet Frame Preamble

- Notifies a host of incoming traffic

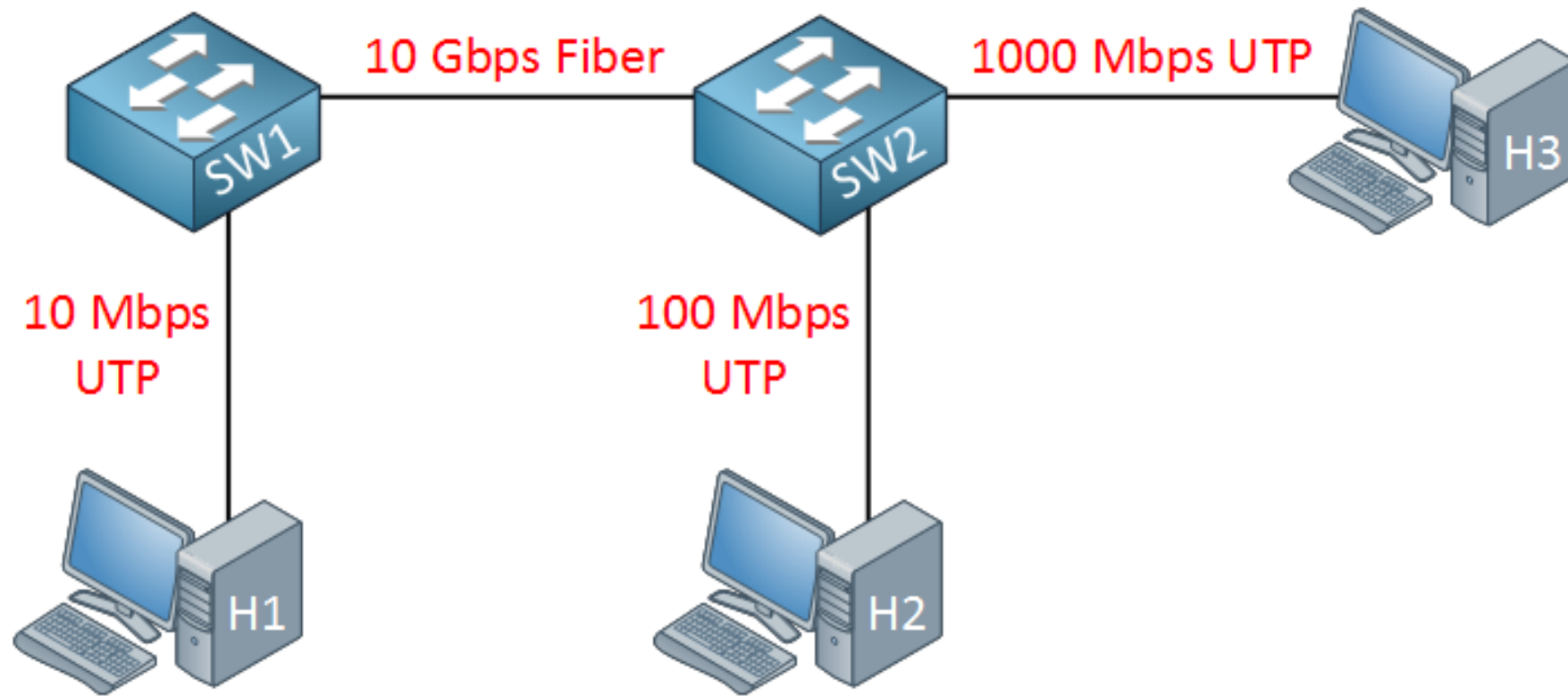
Can be VLAN tagged on trunk links

Carries the Layer 3 payload

Has a CRC check trailer frame



Ethernet Example



Network Contention



What is Network Contention?

Multiple nodes try to use the network at the same time

Contention leads to collisions

Contention needs to be managed



CSMA/CD

Carrier Sense Multiple Access with Collision Detection

The LAN access method used in Ethernet networks

When a device wants to gain access to the network, it checks to see if the network is free

- Network is not free, the device waits a random amount of time before retrying
- Network is free and two devices access the line at exactly the same time, their signals collide and both stop and wait a random amount of time before retrying
- When the line is free that last transmission is resent



Collision Domain



What is a Collision Domain?

A shared network segment where collisions can occur:

- Hub
- Coax bus
- Half-duplex switchport

A *collision* occurs when two devices send a frame at the same time on the same network segment

If a transmitting node detects a collision, it will instantly stop transmitting, wait for a random back-off interval, and retransmit the frame

A collision is the most common cause of frame corruption, runts and giants

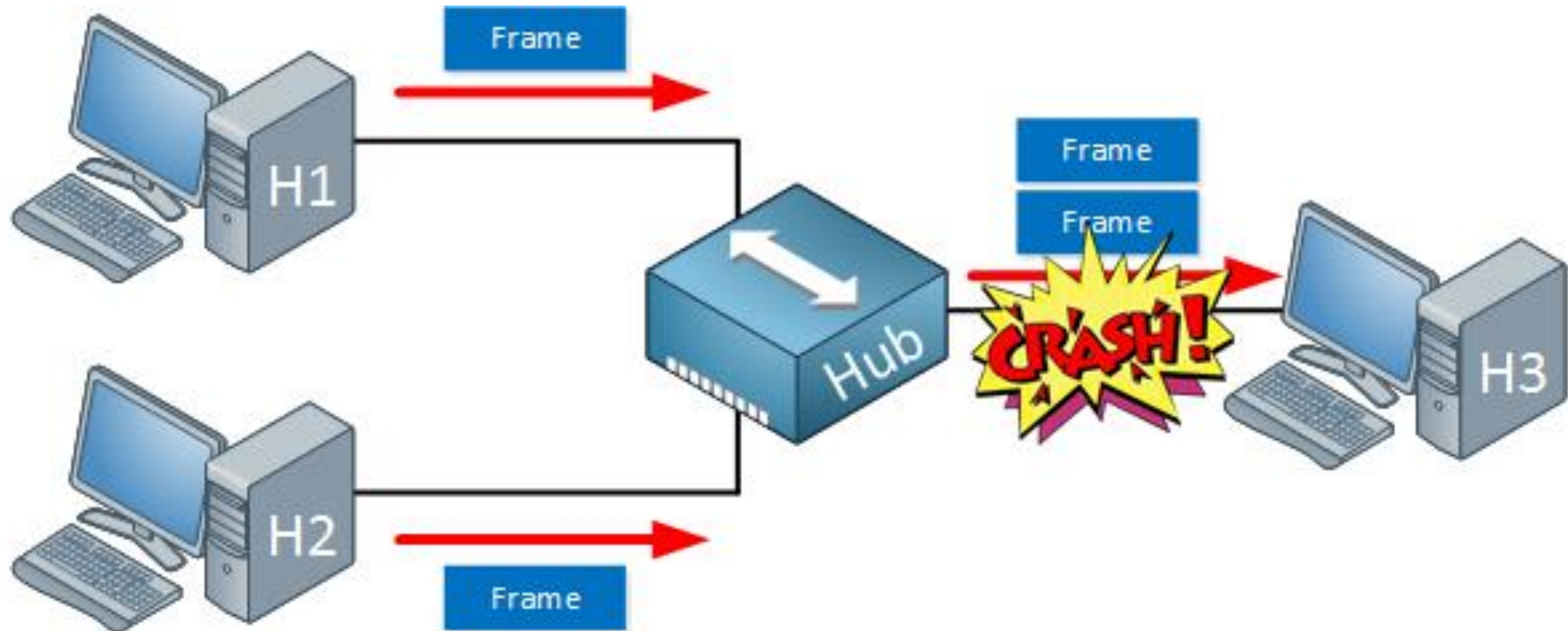
A runt occurs when a transmitting node detects a collision, and stops transmitting a frame before it reaches the minimum size of 64 bytes

A giant is a runt that attaches itself to another frame, creating a frame that exceeds the MTU of 1500 bytes

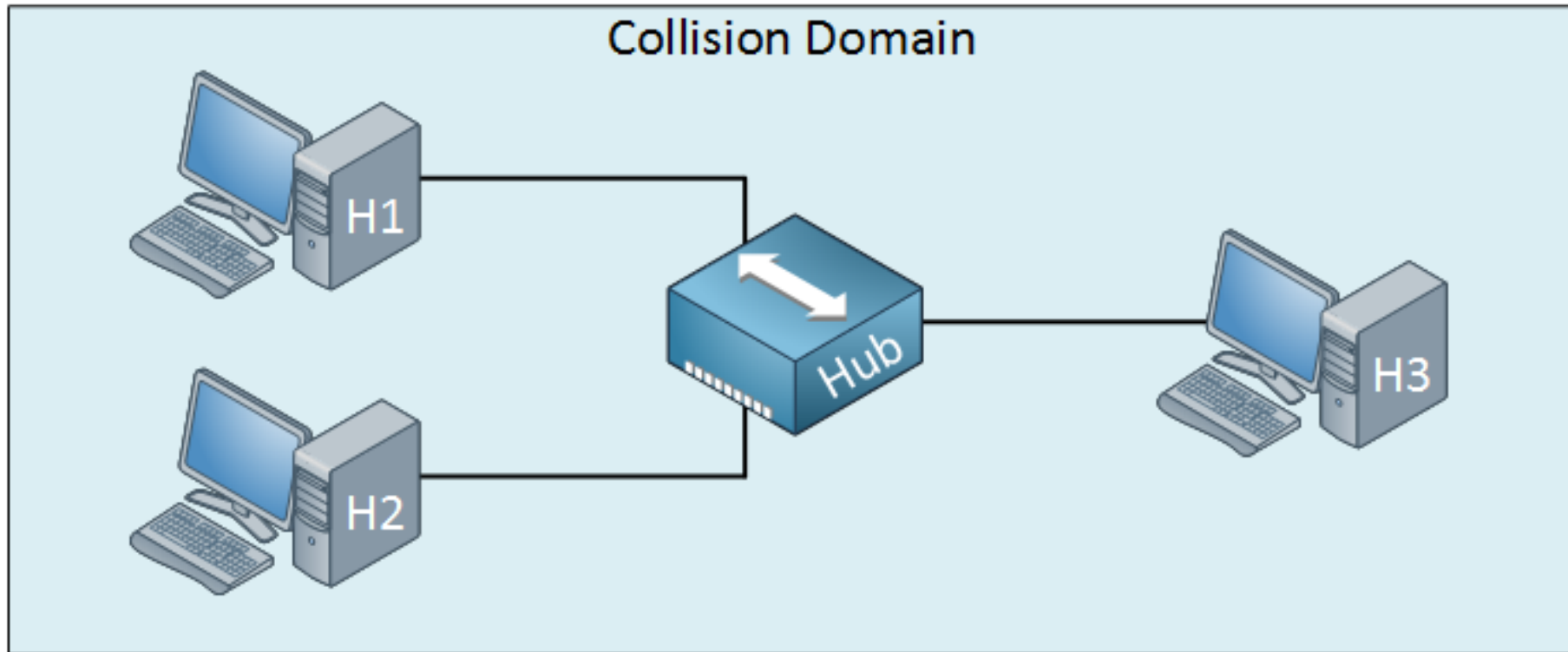
As more nodes are added to a collision domain, the network experiences more and more collisions and becomes less and less efficient



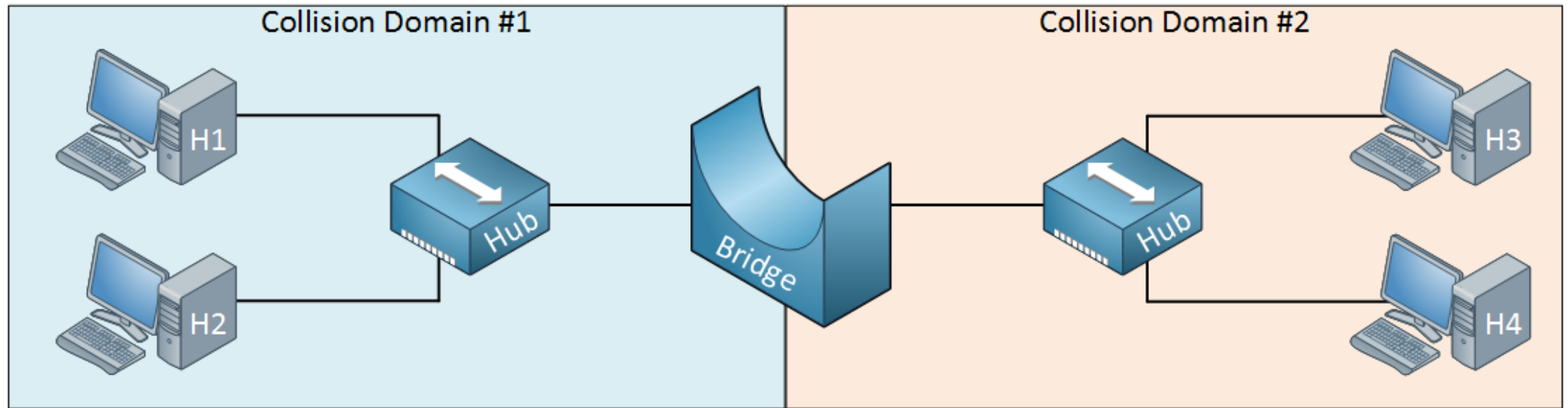
Collision Example



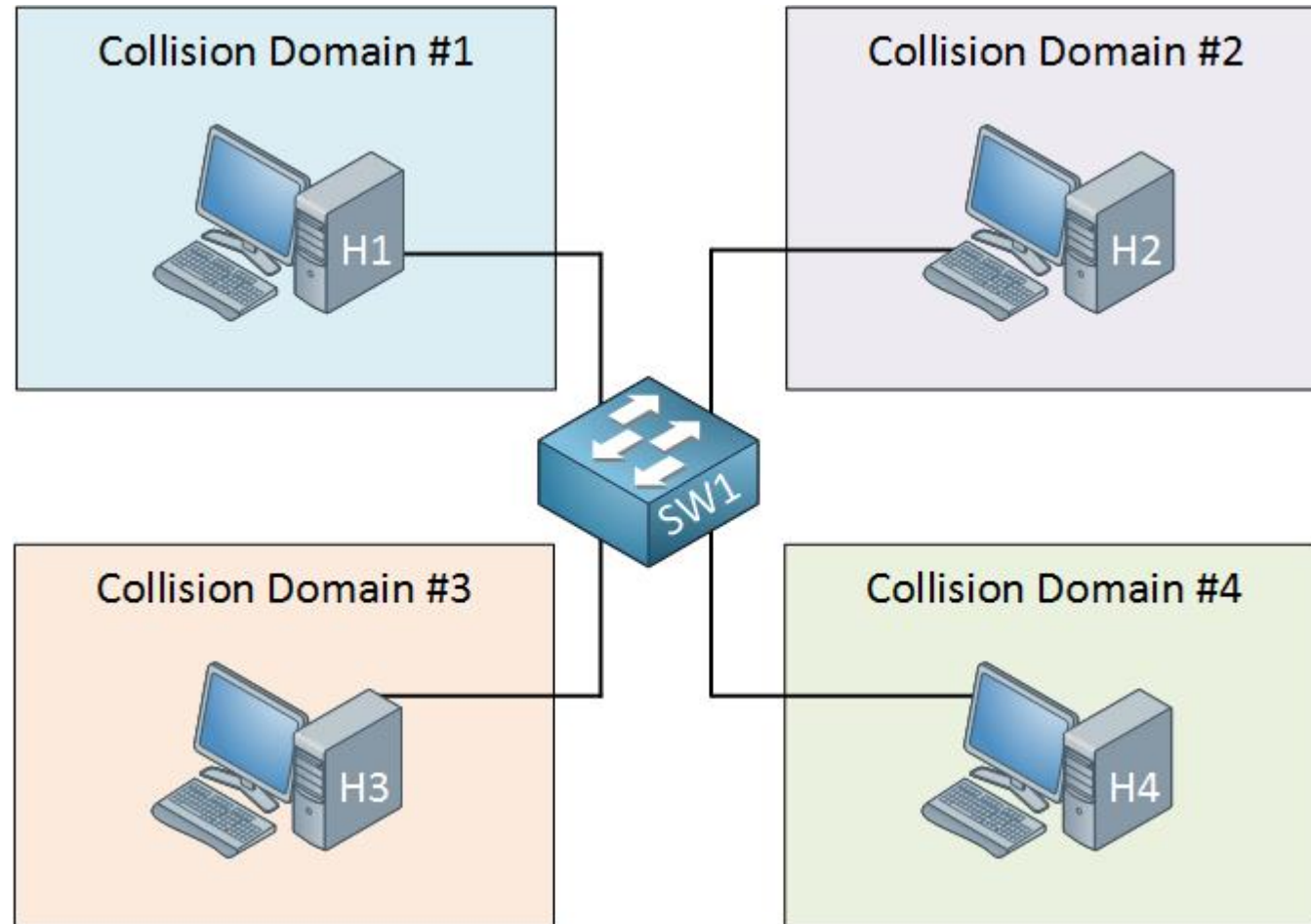
Collision Domain Example



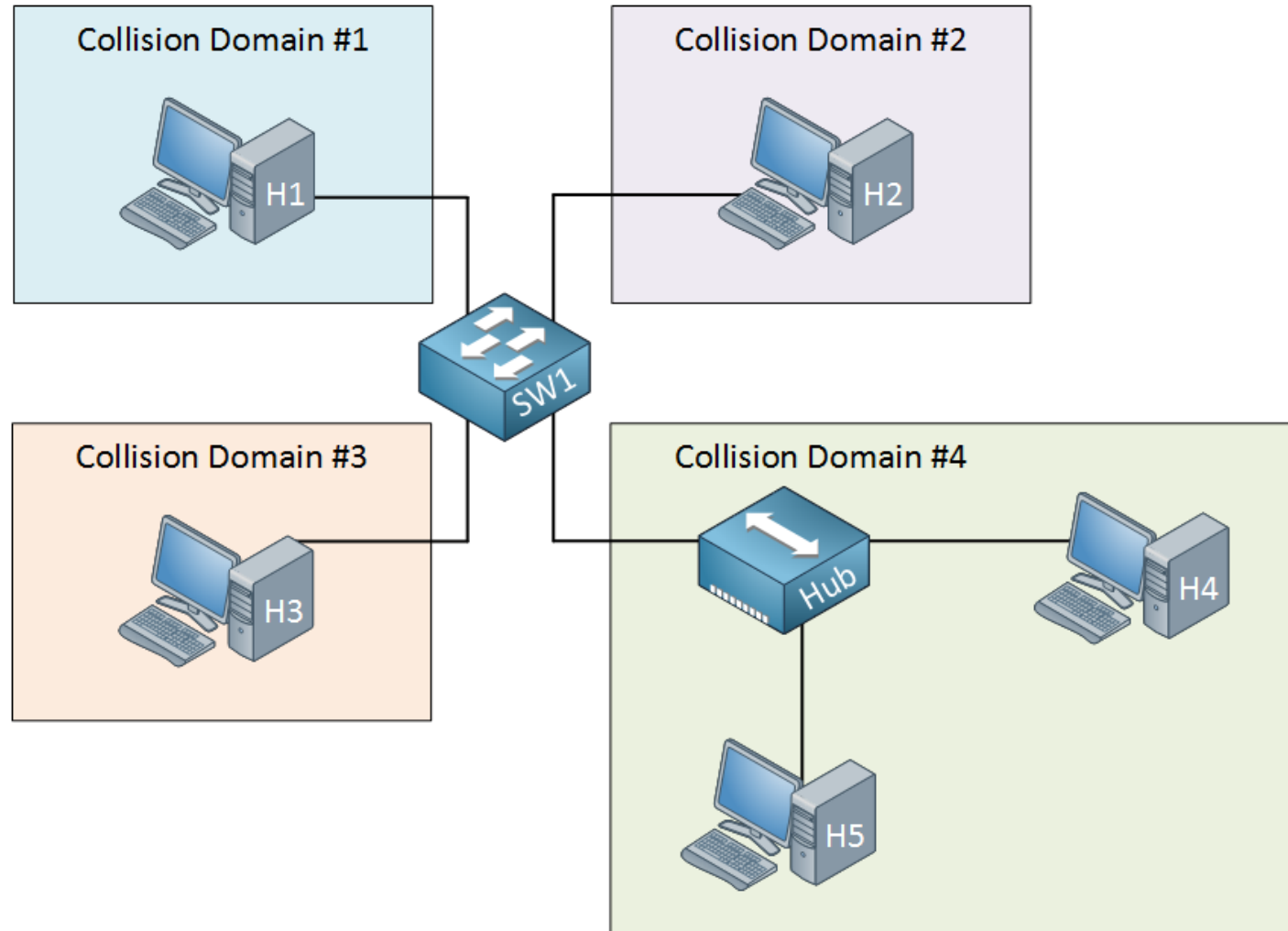
Collision Domain Example #2



Collision Domain Example #3



Collision Domain Example #4



Broadcast Domain



What is a Broadcast Domain?

A network segment where Layer 2 (ARP) broadcasts are allowed to propagate

Can be a physical segment behind a router

Can be a VLAN (even extended across many switches)

Switches separate collision domains

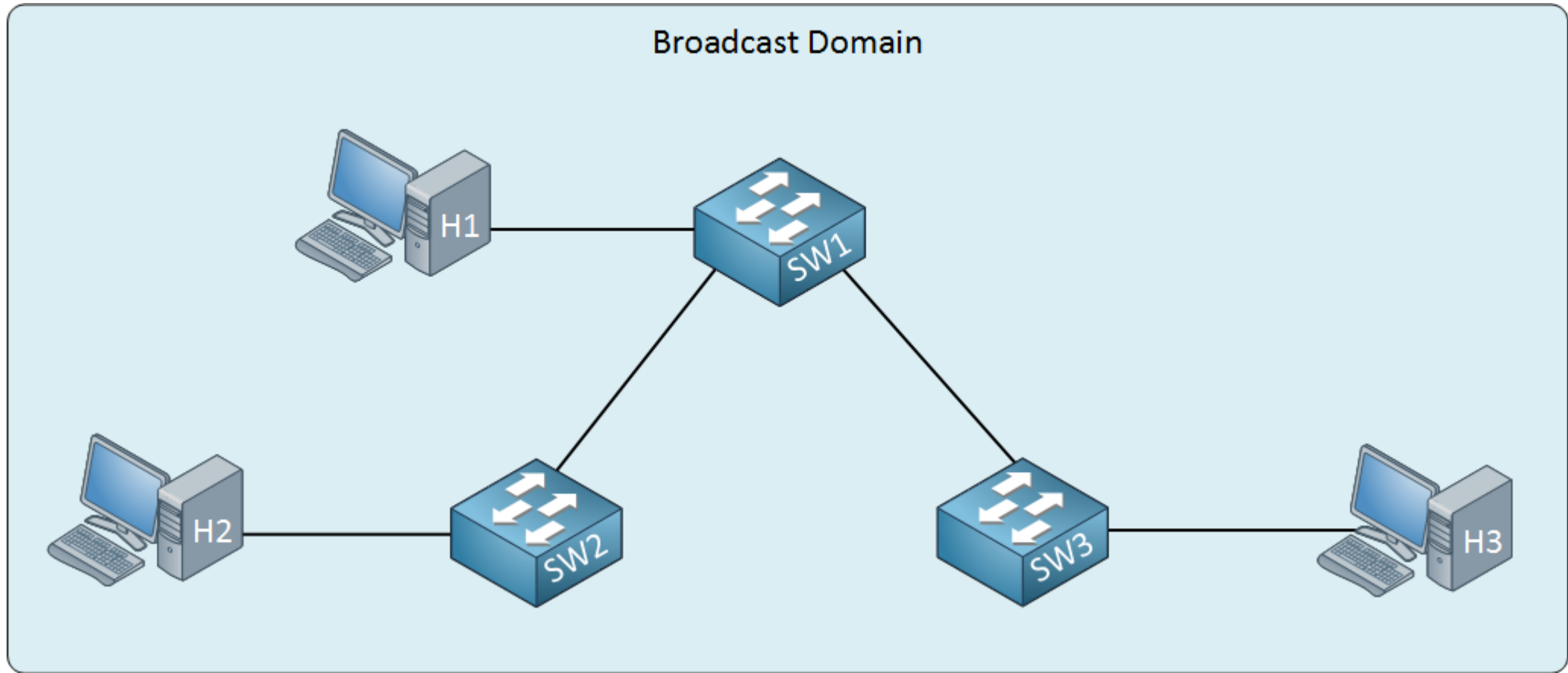
Routers separate broadcast domains

Every router interface will have its own broadcast domain

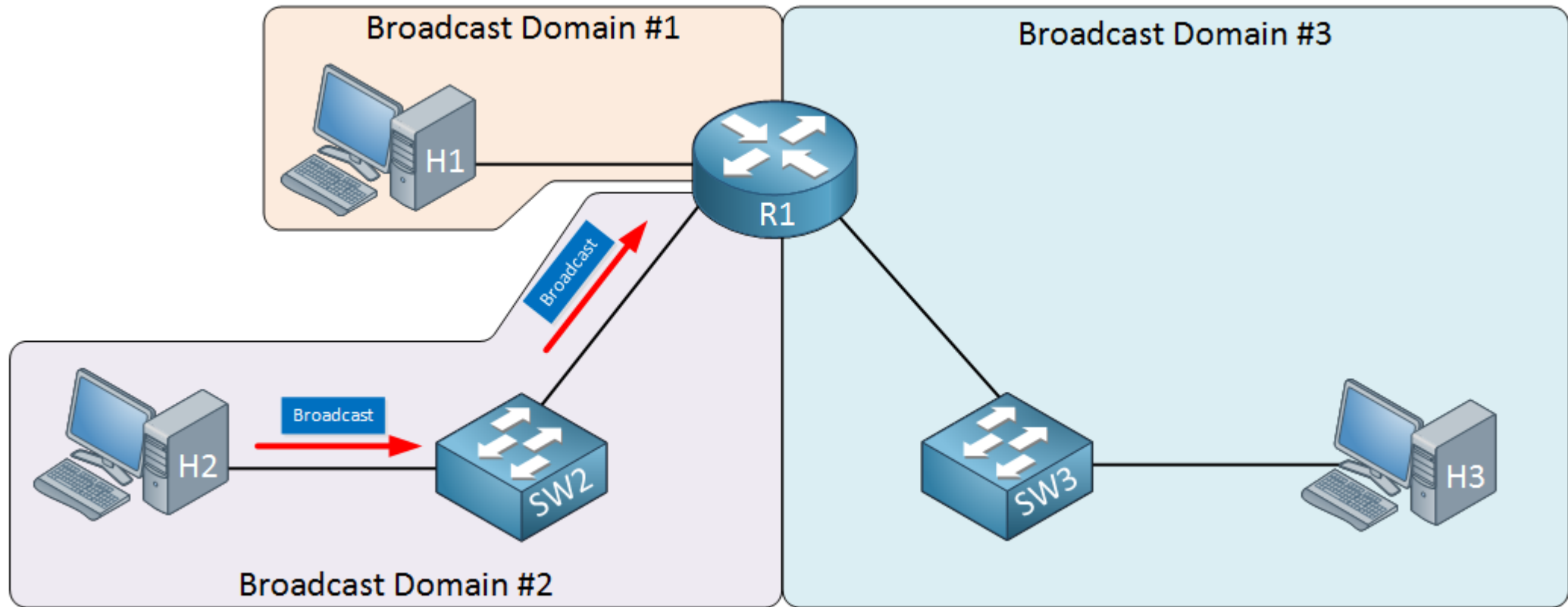
A broadcast domain will also contain at least one collision domain



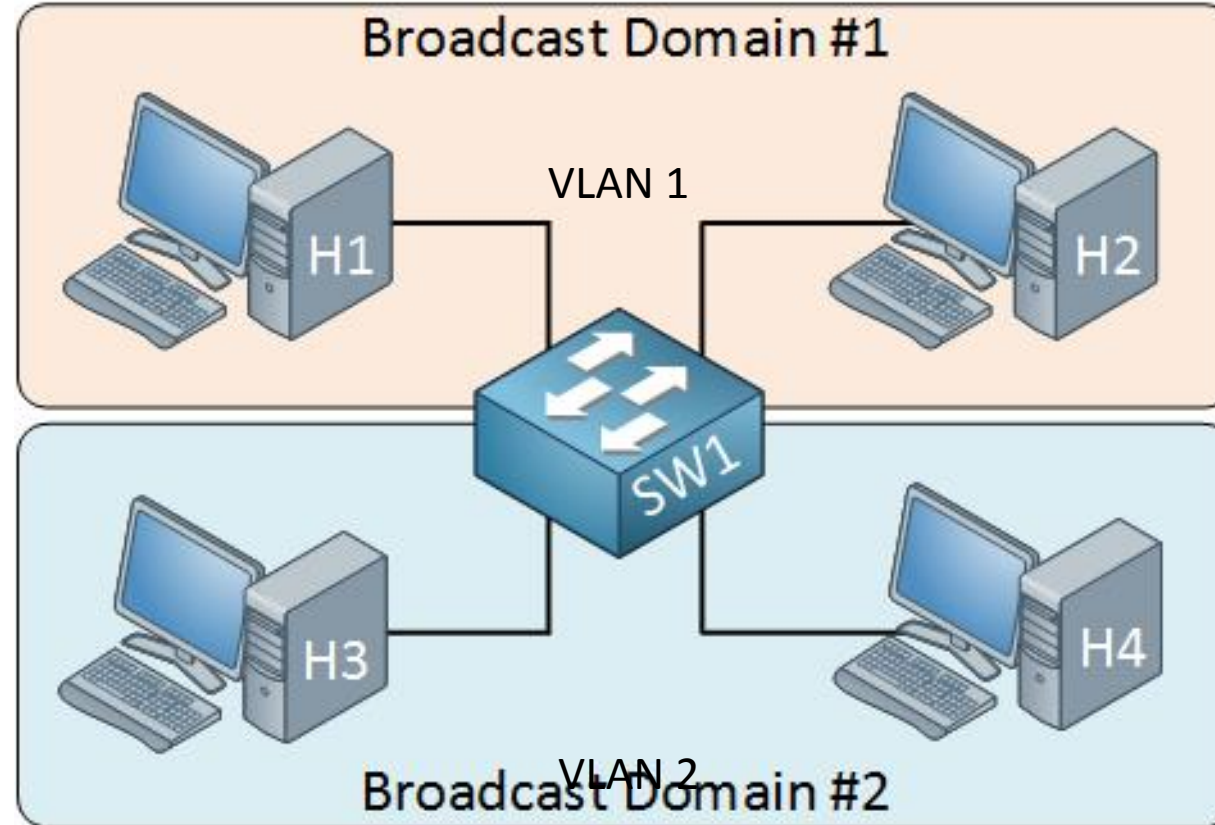
Broadcast Domain Example



Broadcast Domain Example #2



Broadcast Domain Example #3



3.2 Switching Overview

CCNA 200-301 v1.1

Module 3

What is a Switch?

Micro-segmentation

Switching Concepts

Basic Switch Commands

Switch Show Commands

MAC Address Table

MAC Table Commands

Switching Modes

Switching Mode Commands

What is a Switch?

Layer 2 device that makes forwarding decisions based on Layer 2 (MAC) addresses

- Ethernet and Wi-Fi MAC addresses

Learns the MAC address of devices plugged into it

Builds a temporary table (in memory) associating MAC addresses with switchports

Has high port density (many physical ports)

Some models provide Power-over-Ethernet (PoE) on ports for phones, WAPs, cameras, etc.



Micro-segmentation

Switchports in full duplex mode divide a network segment into many micro-segments

- One micro-segment for every switchport

With separate wire pairs for transmit and receive, both sides can talk at the same time with no fear of collisions

A hub, or a switchport in half-duplex mode, is still at risk of collisions

- With a hub, all ports are part of the same collision domain
- With a switch, there are still two nodes (on either side of the link) that can have their traffic collide



Switching Concepts

MAC Address Table

- A temporary table in the switch's memory
- Maps MAC addresses to switchports

MAC Learning

- The process by which a switch populates its MAC address table

MAC Aging

- The process by which entries in the MAC address table are removed after a certain period of inactivity
- Cisco switches by default age out inactive entries after 5 minutes
- Ensures that the MAC address table remains up-to-date and does not become cluttered with stale entries



Switching Concepts (cont'd)

Frame Switching

- Forward incoming frames to the port(s) listed in the MAC table

Frame Flooding

- Broadcasts, multicasts, and unknown unicasts - flood the frame out all ports (including trunk links)
- Except the port it was received on



Basic Switch Commands

Command	Description
<code>enable</code>	Move into privileged mode
<code>configure terminal</code>	Move into global configuration mode
<code>hostname [name]</code> <code>hostname DSW1</code>	Give the switch a name
<code>enable secret [password]</code> <code>enable secret letmein</code>	Require a password to enter privileged mode
<code>interface [type][number]</code> <code>interface g0/7</code>	Move into interface configuration mode
<code>interface range [type] [port-range]</code> <code>Interface range tengigabit 0/1-4</code>	Select a range of interfaces to further apply configurations to as a group



Basic Switch Commands

Command	Description
<code>exit</code>	Exit out of the current mode to the lower mode
<code>end</code>	Jump back to the “top” of privileged mode
<code>shutdown</code>	Administratively shut down the interface
<code>no shutdown</code>	Administratively bring up the interface
<code>copy running-config startup-config</code>	Save the configuration



Switch Show Commands

Command	Description
<code>show version</code>	Display information about the device
<code>show interfaces</code>	Display information about all interfaces on the device
<code>show interfaces [type] [number]</code> <code>show interfaces g0/7</code>	Display information about a particular interface
<code>show interfaces status</code>	Displays a table of interfaces, if they are connected, their VLAN, duplex and speed, interface type
<code>show ip interfaces brief</code>	Displays a table of interfaces, their IP addresses, and their up/down status



Switch Show Commands (cont'd)

Command	Description
<code>show running-config</code>	Displays the current running configuration
<code>show startup-config</code>	Displays the configuration that will load upon reboot
<code>show cdp neighbors</code>	Displays basic information about immediately connected devices (that support CDP)
<code>show cdp neighbors detail</code>	Provides detailed information about CDP neighbors, including their IP address



MAC Address Table



MAC Address Table

A temporary table the switch builds in its memory

Displays the current mapping of MAC addresses to switchports

Switch uses it to make forwarding decisions at Layer 2

Built dynamically

- Whenever devices transmit, the switch notes the source MAC and adds it to the table
- Table is stored in the switch's RAM
- If a hub, uplink, or trunk link is attached to a port, that port will have multiple MAC addresses associated with it

If the destination MAC address is not in the table:

- The switch floods the frame out all ports, including trunk links
- Except the port the frame came in on

Can be cleared manually or by rebooting the switch



MAC Address Table Example

```
Switch#show mac address-table
      Mac Address Table
```

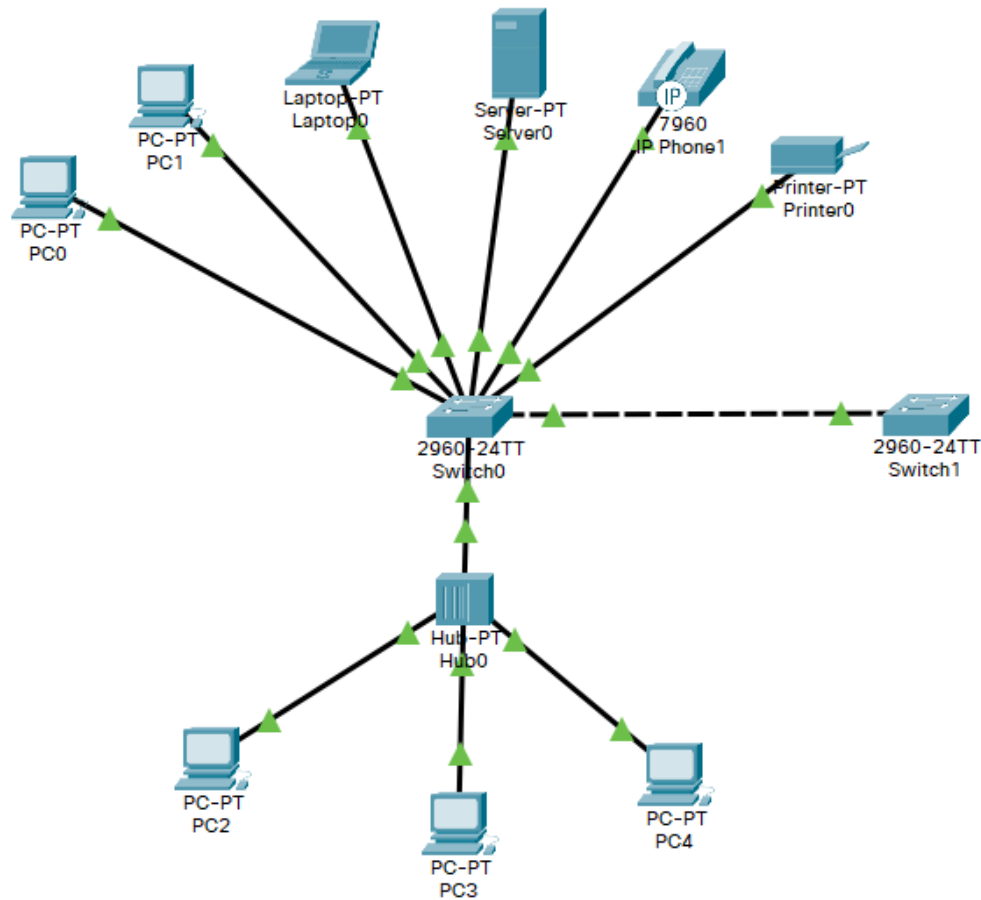
Vlan	Mac Address	Type	Ports
1	0001.4232.5503	DYNAMIC	Fa0/16
1	0001.43c5.e17d	DYNAMIC	Fa0/4
1	0001.634b.c3e3	DYNAMIC	Gig0/1
1	0001.c75a.2a45	DYNAMIC	Fa0/7
1	0001.c789.c29a	DYNAMIC	Fa0/10
1	0001.c937.c0a4	DYNAMIC	Gig0/1
1	0002.162d.1997	DYNAMIC	Fa0/5
1	0002.1761.d2b4	DYNAMIC	Gig0/1
1	0004.9ad7.ac8d	DYNAMIC	Gig0/1
1	0007.ec6a.295c	DYNAMIC	Gig0/1
1	0009.7c75.a43a	DYNAMIC	Fa0/6
1	000a.f36c.1deb	DYNAMIC	Fa0/14
1	000c.8582.619b	DYNAMIC	Gig0/1

Do not mistake a MAC address table for an ARP cache!

- A MAC table maps source MAC addresses to switchports
- An ARP cache maps MAC addresses to destination IP addresses



MAC Address Table Example #2



```
Switch#show mac address-table
Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	0001.6481.1320	DYNAMIC	Fa0/1
1	0001.9727.4996	DYNAMIC	Fa0/6
1	0003.e48b.5296	DYNAMIC	Fa0/4
1	0005.5e43.6c19	DYNAMIC	Gig0/1
1	0010.114b.8ba1	DYNAMIC	Fa0/7
1	0030.a392.9dd0	DYNAMIC	Fa0/7
1	0060.3e0b.e6d7	DYNAMIC	Fa0/3
1	0060.479d.d02e	DYNAMIC	Fa0/2
1	0090.21e7.c128	DYNAMIC	Fa0/7

```
Switch#show mac address-table
Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	0001.6481.1320	DYNAMIC	Gig0/1
1	0003.e48b.5296	DYNAMIC	Gig0/1
1	0010.114b.8ba1	DYNAMIC	Gig0/1
1	0030.a392.9dd0	DYNAMIC	Gig0/1
1	0060.2fc2.d419	DYNAMIC	Gig0/1
1	0060.3e0b.e6d7	DYNAMIC	Gig0/1
1	0060.479d.d02e	DYNAMIC	Gig0/1
1	0090.21e7.c128	DYNAMIC	Gig0/1



MAC Table Commands

Command	Description
<code>show mac address-table</code>	Show the entire MAC address table, including dynamic and static MAC addresses, associated VLANs and ports
<code>show mac address-table dynamic</code>	Show dynamically learned MAC addresses
<code>show mac address-table static</code>	Show statically configured MAC addresses
<code>show mac address-table interface [type][#]</code> <code>show mac address-table interface g0/10</code>	Show MAC addresses on a specific interface
<code>show mac address-table vlan <vlan id></code> <code>show mac address-table vlan 1</code>	Display the MAC addresses associated with a specific VLAN
<code>clear mac address-table dynamic</code>	Clear the dynamically learned MAC addresses from the MAC table



MAC Table Commands (cont'd)

Command	Description
<pre>[no] mac address-table static <MAC address> vlan <vlan id> interface <type><#> mac address-table static aaaa.bbbb.cccc vlan 10 interface f0/10</pre>	• Manually add/remove a static MAC to the MAC address table • Bind it to a specific VLAN and interface
<pre>show mac address-table aging-time</pre>	Display aging time (in seconds) for dynamically learned MAC addresses
<pre>mac address-table aging-time <seconds> vlan <id> mac address-table aging-time 300 vlan 10</pre>	Configure aging time for dynamic MAC addresses on a specific VLAN
<pre>show mac address-table address <MAC address> show mac address-table address aaaa.bbbb.cccc</pre>	Display VLAN and interface associated with a specific MAC



Switching Modes



Switching Modes

Store and Forward

- Buffer the entire frame
- Run a CRC check on frame to make sure it's not damaged
- Discard frame if damaged
- Forward to destination if ok

Fragment Free

- Validate the first 64 bytes of the frame as it comes in
- The first 64 bytes is the most likely time a collision will occur
- If a collision occurs, the colliding nodes will stop transmitting
- But you still have a runt frame on the segment
- If first 64 bytes are ok, forward as the rest of the frame comes into the switch

Cut-through

- Do not check the frame at all
- Immediately start to forward as it comes in
- Trade accuracy for performance



Configuring Switching Modes Example

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# switching-mode fragment-free
```

```
Switch(config)# switching-mode store-and-forward
```

```
Switch(config)# switching-mode cut-through
```

```
Switch(config)# exit
```

```
Switch# copy running-config startup-config
```

```
Switch# show switching-mode
```



Switching Modes Commands

Command	Description
<code>switching-mode fragment-free</code>	Validate the first 64 bytes of the frame as it comes in
<code>switching-mode store-and-forward</code>	Buffer the entire frame before forwarding
<code>switching-mode cut-through</code>	• Do not wait to buffer the frame • Start forwarding immediately

3.3 Spanning-Tree Protocol (STP)

CCNA 200-301 v1.1

Module 3

Switching Loop

Spanning Tree Protocol (STP)

Root Bridge

STP Port Roles

STP Port States

Spanning-Tree Commands

STP Port Configuration Commands

STP Timer Commands

Spanning Tree Show Commands

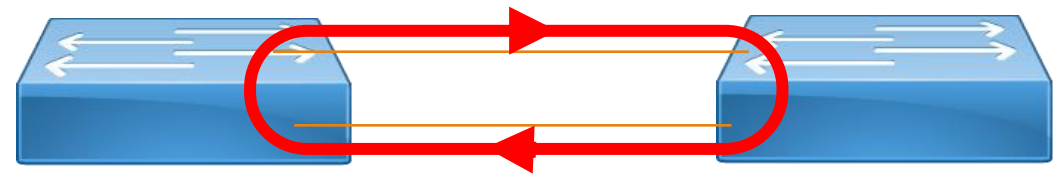
Switching Loop

Occurs on a network when there is a redundant link between switches

Switches by their nature flood broadcasts, multicasts, and unknown unicasts out all ports (except the port it was received on)

Loops cause broadcast storms, creates multiple frame copies, and causes network instability:

1. A host sends a Layer 2 broadcast (such as an ARP or DHCP discover)
2. The switch repeats the broadcast out all other ports, including the redundant link to the other switch
3. The other switch in turn repeats the broadcast out all its ports, including the first link to the first switch
4. The process repeats endlessly
5. Switch CPU utilization shoots up to 80+%
6. Switch MAC tables become unstable



Spanning Tree Protocol (STP)

IEEE 802.1d

Used to eliminate Ethernet switching loops

- Prevents forwarding loops by port blocking

Ensures that there is only one active path between two network devices at any given time

- Eliminates the possibility of a loop



Root Bridge Election

Every switch running STP has a unique Bridge ID

- Priority value + switch's MAC address
- The default priority value on all switches is 32,768

STP begins by electing a "Root Bridge"

- The election process is automatic
- Switch with the lowest Bridge ID wins
- First compare priority values – lowest wins
- In case of tie (quite common) then lowest MAC address wins

You can manually configure a switch to always be selected as the Root Bridge

- A core switch is typically considered a good candidate to be the Root

You can manually configure a secondary root bridge to take over if the primary root bridge fails

- Configure the priority value of the secondary to be slightly higher (worse) than the primary
- Ensure that it is lower (better) than all of the other switches
- Priority values can be configured from 0 to 61,440 in increments of 4096



STP Port States

Blocking

- The port does not forward frames
- It only listens for BPDU (Bridge Protocol Data Unit) messages
- Link stays blocked indefinitely until the topology changes
- If the topology changes, will remain blocking for 20 seconds before recalculating topology
- Link light is amber

Listening

- The port is preparing to forward data but not yet forwarding
- It continues to process BPDUs
- Link remains listening for 15 seconds
- Link light is amber



STP Port States (cont'd)

Learning

- The port begins to learn MAC addresses but still does not forward data frames
- Link remains learning for 15 seconds
- Link light is amber

Forwarding

- The port forwards data frames and learns MAC addresses
- Link forwards indefinitely until topology changes (or switch reboots)
- Link light is green

Disabled

- The port is not participating in STP, meaning it does not forward frames or process BPDUs
- Depending on the switch model, the link light can be amber or off



STP Port Roles

Root Port (RP)

- The port on a non-root switch with the lowest cost path to the Root Bridge
- Each non-root switch has one Root Port

Designated Port (DP)

- The port on a switch that has no redundancy and is forwarding traffic to a specific network segment
- Each segment has one Designated Port

Non-Designated Port

- Ports that are not forwarding frames (to avoid loops) but are still active and listening for STP updates

Note: On a root bridge, all ports are called “Designated Port”
Only non-root bridges have Root Ports



Identifying and Blocking Redundant Links

The Root Bridge acts as a reference point for all path calculations in the network

- The root bridge sends Root Bridge Protocol Data Units (Root BPDUs) out all of its ports
- As downstream switches receive Root BPDUs on their ports, they know that port has a path to the root
- If multiple ports receive Root BPDUs, the switch knows it has a switching loop

Switches update and forward Root BPDUs to their downstream neighbors

- Each switch adds its own link cost to the total cost listed in the BPDU
- The path with the lowest cumulative cost is considered the "best" path to the root bridge
- All other paths are put on standby



Identifying and Blocking Redundant Links (cont'd)

A switch that detects loops will block its own redundant port(s) to break the loop

First based on total path cost to the root for that port

In case of tie, then based on port number

Lowest path cost, then port number, wins!

Losers are placed in a blocking state

The admin can manually assign a port priority to override losing due to higher (less desirable) port number

- Often done when the winning port has an unstable link



STP Example

Switch0

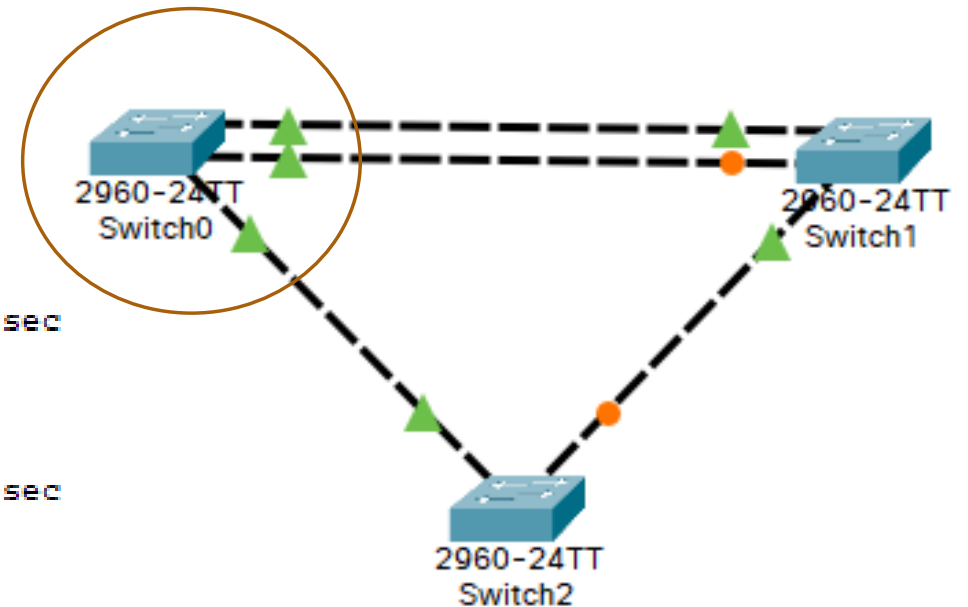
```
Switch#show spanning-tree
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    32769
           Address    000C.CF08.5A64
           This bridge is the root
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
           Address    000C.CF08.5A64
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Gi0/1	Desg	FWD	4	128.25	P2p
Gi0/2	Desg	FWD	4	128.26	P2p



STP Example (cont'd)

Switch1

```
Switch#show spanning-tree
```

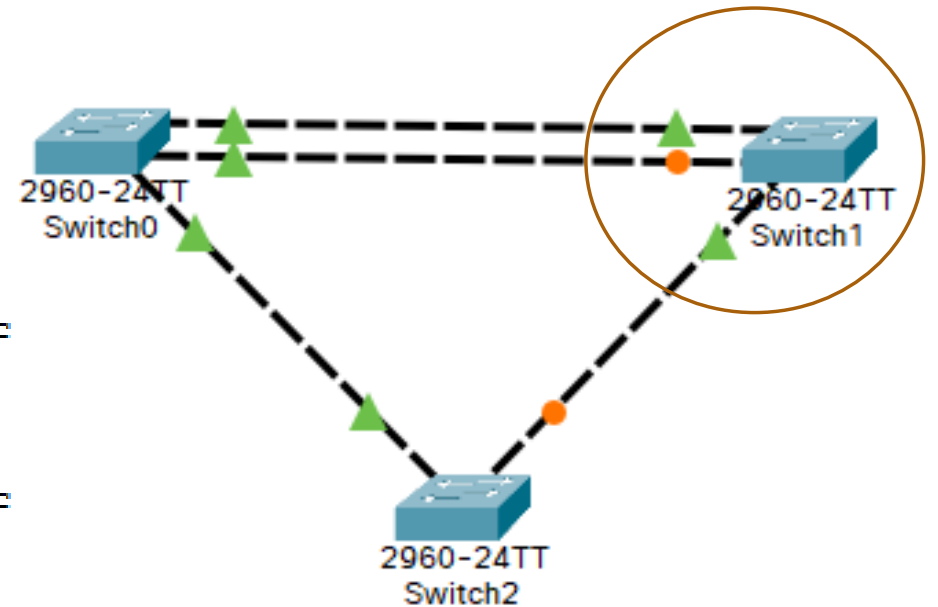
```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    32769
Address     000C.CF08.5A64
Cost        4
Port        25(GigabitEthernet0/1)
Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
```

```
Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
Address      00E0.8F8E.C082
Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
Aging Time   20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/2	Altn	BLK	4	128.26	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Gi0/1	Root	FWD	4	128.25	P2p



STP Example (cont'd)

Switch2

```
Switch#show spanning-tree
```

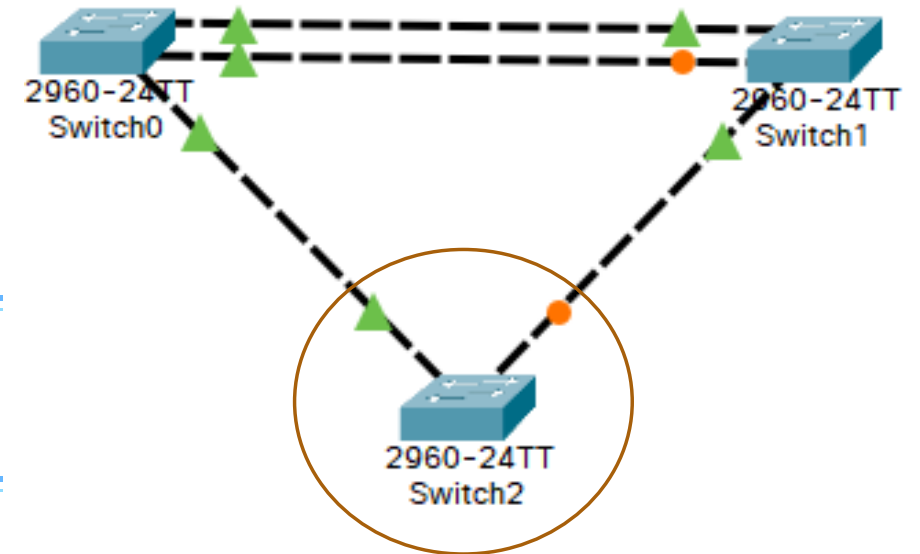
```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID      Priority    32769
Address      000C.CF08.5A64
Cost         19
Port         1(FastEthernet0/1)
Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID    Priority    32769  (priority 32768 sys-id-ext 1)
Address      00D0.BCC1.E26A
Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
Aging Time    20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p



Spanning-Tree Commands

Command	Description
<code>spanning-tree</code>	• Enable spanning-tree on the switch • It is enabled by default
<code>show spanning-tree</code>	Display STP information for the switch
<code>show spanning-tree vlan <vlan-id></code> <code>show spanning-tree vlan 100</code>	If you want to see STP details for a specific VLAN
<code>spanning-tree mode <pvst rapid-pvst mst></code> <code>spanning-tree mode rapid-pvst</code>	Change STP to one of its variant modes rapid-pvst (Rapid PVST+) is the preferred these days
<code>spanning-tree vlan <vlan-id> priority <value></code> <code>spanning-tree vlan 20 priority 4096</code>	• Influence the root bridge election process • Priority can be 0 – 61440 in increments of 4096 • Lowest is most likely to win
<code>spanning-tree portfast</code>	Put the switchport into portfast mode
<code>spanning-tree portfast default</code>	Enable portfast on all ports globally



STP Port Configuration Commands

Command	Description
<code>spanning-tree bpduguard enable</code>	Prevent a PortFast-enabled port from sending or receiving BPDUs – no guardrails against a loop
<code>spanning-tree portfast bpduguard default</code>	Enable BPDUGuard globally on all access ports
<code>spanning-tree bpduguard enable</code>	Disable a specific PortFast-enabled port if it receives a BPDU – put port in err-disable state
<code>spanning-tree portfast bpduguard default</code>	Enable BPDUGuard globally for all access ports
<code>spanning-tree vlan <vlan-id> root primary</code> <code>spanning-tree vlan 3 root primary</code>	Manually set a switch to bridge priority be 24577. If other stay at default, it becomes the root.
<code>spanning-tree vlan <vlan-id> root secondary</code> <code>spanning-tree vlan 3 root secondary</code>	• Manually set a switch to be a secondary root • No strict limit – but best to only have one



Spanning Tree Timer Commands

Command	Description
<code>spanning-tree vlan <vlan-id> hello-time <seconds></code> <code>spanning-tree vlan 50 hello-time 3</code>	Adjust hello-time timers From 1 – 10 seconds Default is 2 seconds
<code>spanning-tree vlan 50 forward-time <seconds></code>	• How long a port stays in the listening and learning states before transitioning to the forwarding state • From 4 – 30 seconds, split evenly between listening and learning states
<code>spanning-tree vlan 50 max-age <seconds></code>	• Set how long a switch will waiting before aging out BPDU info if no new BPDU is received • From 6 – 40 seconds • Default is 20



Spanning Tree Show Commands

Command	Description
<code>show spanning-tree</code>	Provide details about the root bridge, local bridge, and port roles for all VLANs
<code>show spanning-tree summary</code>	Display summary of spanning tree status across all VLANs
<code>show spanning-tree vlan <vlan-id></code> <code>show spanning-tree vlan 1</code>	Display Spanning Tree information for a specific VLAN
<code>show spanning-tree interface <interface-id></code> <code>show spanning-tree interface fa0/9</code>	Display Spanning Tree information for a specific interface



Show Spanning Tree Output Example

The switch that
won the root
election

The switch you're
currently on

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
 Address 000C.85FA.4C00
 Cost 4
 Port 1 (GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 000D.BDFA.2E80
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/1	Root	FWD	4	128.1	P2p
Gi0/2	Desg	FWD	4	128.2	P2p
Gi0/3	Altn	BLK	4	128.3	P2p



3.4 Rapid PVST+

CCNA 200-301 v1.1

Module 3

Rapid PVST+ (Per-VLAN Spanning Tree Plus)

Rapid PVST+ Features

Rapid PVST+ Port States

Rapid PVST+ Port Roles

Rapid PVST+ Commands

Rapid PVST+ Port Configuration Commands

Rapid PVST+ Show Commands

Rapid PVST+ Timer Commands

Rapid PVST+ (Per-VLAN Spanning Tree Plus)

Cisco's enhancement to the IEEE 802.1w Rapid Spanning Tree Protocol (RSTP)

Rapid PVST+ provides faster convergence times than the original STP

- 15 seconds or less total

Each VLAN has its own root bridge and spanning-tree topology



Rapid PVST+ Features

Rapid PVST+ automatically treats ports configured with PortFast as edge ports

- These ports are considered to be directly connected to end devices
- Are immediately placed in the forwarding state, bypassing the usual STP states

Rapid PVST+ BPDUs are sent at regular intervals by all switches, not just the root bridge

- This allows faster detection of topology changes and quicker convergence
- Point-to-point full-duplex links are allowed to transition faster than half-duplex links

If a switch running Rapid PVST+ detects a legacy STP BPDU it falls back to standard STP for that VLAN

If switch Rapid PVST+ timers are mismatched, they will synchronize with the Root Bridge



Rapid PVST+ Port States

The port states in Rapid PVST+ are simplified compared to traditional STP

Discarding

- Combines the blocking and listening states from traditional STP
- Ports do not forward frames and do not learn MAC addresses

Learning

- The port learns MAC addresses for a few seconds but does not forward frames

Forwarding

- The port forwards frames and continues to learn MAC addresses



Rapid PVST+ Port Roles

Root port

- The port on a non-root bridge that has the overall best path cost to the root bridge

Designated port

- A port designated to forward packets on a given network segment or link
- Has the lowest path cost to the root bridge for that segment

Alternate port

- A backup port that provides an alternate path to the root bridge
- Offers a different path to reach the root bridge in case the current root port fails

Backup port

- A port that provides a redundant connection to a segment where another port on the same switch is already serving as the designated port
- Mainly found in scenarios with redundant links to the same network segment

Disabled port

- Not strictly an STP port role
- The port has been administratively shut
- It plays no role in STP and has no impact on STP topology



Rapid PVST+ Commands

Command	Description
<code>spanning-tree mode rapid-pvst</code>	Enable Rapid PVST+ on the switch
<code>show spanning-tree</code>	Display configuration, including mode (Rapid PVST+), root bridge for each VLAN, and port roles
<code>show spanning-tree vlan <vlan-id></code> <code>show spanning-tree vlan 15</code>	Shows STP details for a specific VLAN, including the root bridge, timers, and port roles
<code>spanning-tree portfast</code>	Immediately transition the port to forwarding state
<code>spanning-tree portfast default</code>	Apply PortFast to all access ports



Rapid PVST+ Port Configuration Commands

Command	Description
<code>spanning-tree bpdupfilter enable</code> <code>spanning-tree portfast bpdupfilter default</code>	• Prevent a PortFast-enabled port from sending or receiving BPDUs – at risk of loops • Enable it globally for all access ports
<code>spanning-tree bpduguard enable</code> <code>spanning-tree portfast bpduguard default</code>	• Prevent loops by disabling a PortFast-enabled port if it receives a BPDU – put port in blocking err-disable state • Enable it globally for all access ports



Rapid PVST+ Port Configuration Commands (cont'd)

Command	Description
<pre>spanning-tree vlan <vlan-id> priority <value> spanning-tree vlan 3 priority 4096</pre>	• Lowers switch priority for a specific VLAN • Increase its chances of becoming the root • Can range from 0 (highest priority) to 61440 (lowest) in increments of 4096
<pre>spanning-tree vlan <vlan-id> root primary spanning-tree vlan 3 root primary</pre>	Manually set a switch to be the root bridge
<pre>spanning-tree vlan <vlan-id> root secondary spanning-tree vlan 3 root secondary</pre>	Manually set a switch to be a secondary root



Rapid PVST+ Show Commands

Command	Description
<code>show spanning-tree</code>	Display detailed information about the root bridge, local bridge, and the state of each port in the spanning tree topology
<code>show spanning-tree interface <interface-id></code> <code>show spanning-tree interface fa0/10</code>	Display detailed STP information for a specific interface, including its current state (Forwarding, Learning, etc.), role, and cost
<code>show spanning-tree summary</code>	Provides a summary of the STP configuration, including BPDU Guard, PortFast, and other global setting



Rapid PVST+ Timer Commands

Command	Description
<code>spanning-tree vlan <vlan-id> hello-time <seconds></code> <code>spanning-tree vlan 1 hello-time 1</code>	• Adjust hello-time timers • 1 – 10 seconds • Default 2
<code>spanning-tree vlan <vlan-id> forward-time <seconds></code> <code>spanning-tree vlan 1 forward-time 10</code>	• Adjust forward-delay timers • 4 – 30 seconds • Default 15
<code>spanning-tree vlan <vlan-id> max-age <seconds></code> <code>spanning-tree vlan 1 max-age 15</code>	• Adjust max-age timers • 6 – 40 seconds • Default 20

Note: In Rapid PVST+ you cannot set timers globally for all VLANs on the switch

3.5 VLANs

CCNA 200-301 v1.1

Module 3

Virtual LAN (VLAN)

Configuring VLANs

Access Port Types

Default VLANs

VLAN Commands

VLAN Show Commands

Virtual LAN (VLAN)

A VLAN is a logical grouping of access switchports

- Ports need not be physically next to each other
- All modern switches support 1 – 1005 VLANs
- Some support up to 4096 VLANs

Used to group devices for management or security purposes

- Devices plugged into a switchport have no awareness of being in a VLAN

VLANs can be extended across switches via trunk links

Ports in a VLAN can only communicate with ports in the same VLAN

- A VLAN is a broadcast domain

Each VLAN should be assigned its own subnet ID

- Nodes in the VLAN should be configured for that subnet
- VLANs cannot talk to each other unless traffic is routed between VLANs by a router
- Create a VLAN interface on the distribution switch to be the default gateway for all nodes on that VLAN



Access Port Types

An access port on a switch can be one of two types:

Data

- For PCs

Voice

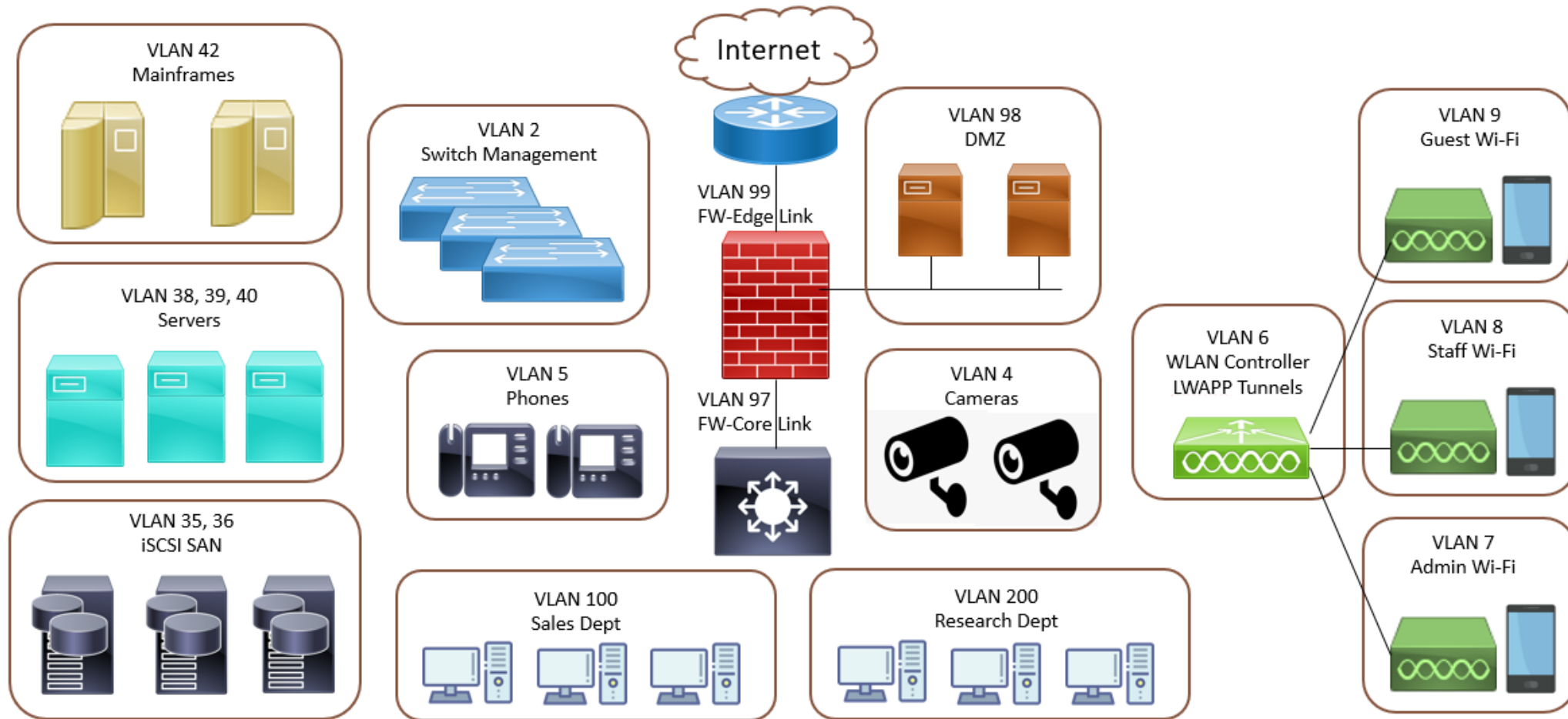
- For IP phones

You can have both a PC and a phone on the same access port

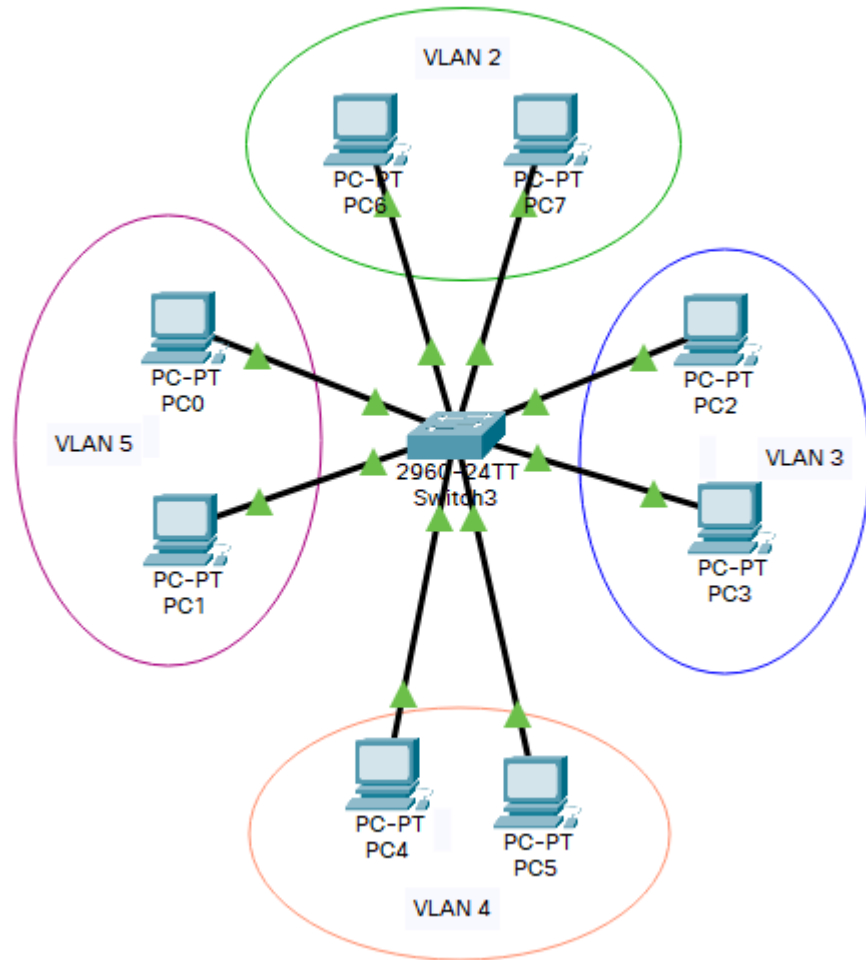
- In most cases, the PC plugs into the phone, and the phone plugs into the switchport
- You will need to specify two VLANs on that port, one for the PC and one for the phone
- Even if only a phone is plugged into the switchport, consider specifying that the port is for voice
 - Allows you to configure Quality-of-Service for the phone



Common VLAN Usage



VLAN Example



VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/4, Fa0/5, Fa0/6 Fa0/8, Fa0/9, Fa0/10, Fa0/12 Fa0/13, Fa0/14, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/23, Fa0/24, Gig0/1, Gig0/2
2	VLAN0002	active	Fa0/7, Fa0/15
3	VLAN0003	active	Fa0/2, Fa0/11
4	VLAN0004	active	Fa0/3, Fa0/22
5	VLAN0005	active	



Configuring VLANs

Create the VLAN on the switch, then add physical ports to the VLAN

- The switchports in one VLAN do NOT need to be physically near each other
- You could have switchports in other rooms or buildings on the LAN belong to the same VLAN
- You can easily change a port's VLAN membership

Switch ports generally only belong to one VLAN at a time

- Exception: when an IP phone connects to the network through a PC
- Switch port has one VLAN for the phone, one VLAN for the PC (data)

Initially, all switch ports are in the same default VLAN (usually VLAN 1)

- If you configure a port to join a particular VLAN and then unjoin the port, it reverts back to the default VLAN
- If you delete the VLAN without unjoining the ports from it, those ports become “orphaned” and (usually) go into a blocking state (stops forwarding traffic)



Default VLANs

VLAN 1

- The default VLAN for all ports on a Cisco switch
- If you don't assign a port to a specific VLAN, it belongs to VLAN 1 by default
- Cannot be deleted
- Typically used for management traffic
- Best practice: change the management VLAN to something else to increase security

VLAN 1002 to 1005

- Default VLANs for Legacy Technologies (for FDDI, Token Ring Default)



Default VLANs (cont'd)

VLAN 1006 to 4094

- Extended VLAN Range
- Supported on certain switch models

VLAN 4095

- Reserved as a system VLAN
- Not a user-configurable VLAN
- Cannot be used for normal network traffic

VLAN 0

- Used for priority tagging in IEEE 802.1p
- Not user-configurable



VLAN Configuration Example

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Marketing
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# interface FastEthernet 0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 10
```

```
Switch# show vlan
```

```
Switch# show vlan brief
```

```
Switch# show interfaces FastEthernet 0/1 switchport
```

```
Switch# show vlan id 10
```



VLAN Commands

Command	Description
vlan [vlan-id] vlan 10	Create a VLAN
no vlan [vlan-id]	Delete a VLAN from the VLAN database
name [vlan-name] name Marketing	Name a VLAN
switchport mode access	Turn a switchport into an access port
switchport access vlan [vlan-id] switchport access vlan 10	Assign an access VLAN to a port



VLAN Show Commands

Command	Description
show vlan	Display VLAN database
show vlan brief	Display VLAN information
show interfaces [interface-id] switchport show interfaces FastEthernet 0/1 switchport	Check VLAN assignment to interfaces
show vlan id [vlan-id] show vlan id 10	Check VLAN membership on a specific port

3.6 VLAN Trunking

CCNA 200-301 v1.1

Module 3

Interswitch Connectivity

VLAN Trunking

VLAN Tagging

802.1Q VLAN Frame Structure

Trunk Link Native VLAN

Native VLAN Mismatch

Trunk Port Negotiation Modes

Trunk Link Negotiation Commands

VLAN Pruning

VLAN Pruning Commands

Interswitch Connectivity

Uplink

- A cable connecting two switches
- Used to add more devices to a single switch port
- All devices belong to the same VLAN as the switchport the uplink is plugged into
- Requires no special configuration

Trunk Link

- An uplink that has been configured to carry traffic from all VLANs from one switch to the next
- A trunk port tags outgoing frames with their respective VLAN ID
- Trunk links extend VLANs across multiple switches
- Requires ports on both sides of the link to be configured as trunk ports



VLAN Trunking

Used to extend VLANs to other switches

- Trunk links carry traffic from all VLANs from one switch to another

A broadcast in a VLAN will extend across the trunk link to all switches and their ports that use that VLAN

IEEE 802.1Q (“dot 1 q”) is the de facto VLAN trunking protocol today

- Older switch models also supported the Cisco proprietary Inter-Switch Link (ISL) trunking protocol



VLAN Tagging

A tag inserted into the Ethernet header identifies which VLAN a particular frame belongs to

- The sending switch applies the VLAN tag to the frame before transmitting it
- The receiving switch then knows which VLAN that frame belongs to
- Tags are only meaningful on a trunk link

Ports that computers, phones, and end devices are plugged into do not tag their frames

- The end devices have no knowledge of the VLAN
- Ports meant for end devices are configured as “access” ports
- Access ports do not use VLAN tags

Default VLAN 1 traffic is untagged (per 802.1q)

- This makes it possible to connect the two switches using a multi-access device such as a hub
- The hub (and any other devices plugged into it) treat the switch traffic as normal Ethernet
- They ignore any VLAN tagging



802.1Q VLAN Frame Structure

802.1Q VLAN frames are distinguished from ordinary Ethernet frames

- 4-byte VLAN tag is inserted into the Ethernet header

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18 ...
Destination address						Source address						VLAN tag				EtherType		Payload
												0x8100		TCI				



Trunk Link Native VLAN

Native VLAN on a trunk link is the VLAN that carries untagged traffic

- Ensures compatibility with devices that do not support VLAN tagging

Frames that are not tagged with any VLAN ID are assumed to belong to the native VLAN

Useful for backward compatibility with older devices that may not support VLAN tagging

- Also handles management traffic or any other untagged traffic that might traverse the trunk link

Can be a potential security vulnerability if not properly managed

- VLAN hopping attacks exploit this by sending double-tagged packets that are stripped of one tag and misrouted

By default is VLAN 1

- Can be changed to any other VLAN
- The native VLAN is configured per trunk port



Native VLAN Mismatch

Occurs if the native VLAN configuration does not match on both ends of a trunk link

- Can also occur if a simple uplink is plugged between two switchports that belong to different VLANs
 - One switchport is in the native VLAN, the other switchport is assigned to a different VLAN

Will lead to errors and traffic being dropped or misrouted

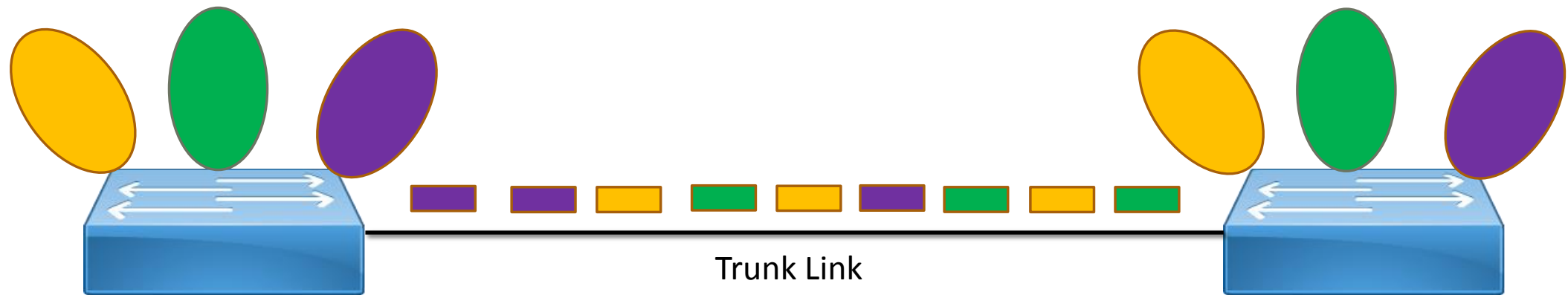
- Spanning-tree will block the port

Best practice is to change the Native VLAN

- Make sure all switches have it set as the global default



Trunk Link Example



Trunk Port Negotiation Modes

Command	Description
switchport mode access	• Forces the interface to be an access port • No attempt to negotiate a trunk link
switchport mode trunk	• Forces the interface to become a trunk, regardless of the configuration on the other end of the link • The port actively attempts to establish a trunk link • If the other side is set to dynamic auto, the link will become a trunk
switchport mode dynamic desirable	• Actively attempts to convert the link to a trunk • Will form a trunk link if the other end is set to trunk, dynamic desirable, or dynamic auto
switchport nonegotiate	• The interface will not participate in negotiation • Use when you want to manually configure the trunk link



Configuring a Trunk Link Example

Do this on both switches:

```
Switch(config)# interface gigabitEthernet 0/1
```

Optional – only necessary on switches such as the Catalyst 3560 that also support ISL

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# end
```

```
Switch# show interfaces trunk
```



Trunk Link Negotiation Commands

Command	Description
<code>switchport mode access</code>	Set the port to be a Layer 2 access port only
<code>switchport trunk encapsulation dot1q</code>	Necessary on some switches before issuing the <code>switchport mode trunk</code> command
<code>switchport mode trunk</code>	Set the port to be a trunk port Actively negotiate to form a trunk link
<code>switchport trunk native vlan <vlan ID></code>	Set the native VLAN on the trunk port
<code>switchport nonegotiate</code>	Set the port to not negotiate a trunk link Assumes the trunk will be manually configured
<code>switchport mode dynamic desirable</code>	Set the port to actively negotiate a trunk link The trunk link forms if negotiation is successful
<code>switchport mode dynamic auto</code>	Set the port to not negotiate, but be willing to become a trunk link if the other side negotiates
<code>show interfaces trunk</code>	Display any trunk links and their status



VLAN Pruning



VLAN Pruning

By default, a trunk link carries traffic from all VLANs across it, even if the other side doesn't have any ports assigned to certain VLANs

- Includes ARP messages and other broadcast / multicast traffic

You can configure a trunk port to “prune” specific VLANs from going in or out of a trunk link

- Will not send or receive traffic from those VLANs
- Helps reduce unnecessary bandwidth consumption on the link and on the switch



VLAN Pruning Example

```
Switch# configure terminal
Switch(config)# interface g0/1
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# switchport trunk allowed vlan remove 30
Switch(config-if)# switchport trunk allowed vlan add 30
Switch(config-if)# end
Switch# copy running-config startup-config
Switch# show interfaces trunk
```



VLAN Pruning Commands

Command	Description
<code>switchport trunk allowed vlan <id, id,...></code> <code>switchport trunk allowed vlan 10,20,30</code>	List the allowed VLANs All others are pruned out
<code>switchport trunk allowed vlan remove <id></code> <code>switchport trunk allowed vlan remove 30</code>	Prune out an existing allowed VLAN
<code>switchport trunk allowed vlan add <id></code> <code>switchport trunk allowed vlan add 40</code>	Add a specific VLAN to the allow list

3.7 VLAN Trunking Protocol (VTP)

CCNA 200-301 v1.1

Module 3

VLAN Trunking Protocol (VTP)

VTP Switch Modes

VTP Commands

VLAN Trunking Protocol (VTP)

A Cisco-proprietary protocol used to manage VLAN configurations across multiple switches within a network

Used by switches to inform each other of existing VLANs

VTP Domain

- A group of switches that share VLAN information
- All switches in the group must have the same VTP domain name

VTP Password

- Optional security enhancement
- If used, must be the same on all switches in a VTP domain



VTP Switch Modes

Server

- The switch can create, modify, and delete VLANs
- Any changes are propagated to all other switches in the same VTP domain
- Will learn from both server and client switches

Client

- The switch cannot create, modify, or delete VLANs
- Instead, it receives VLAN configuration updates from VTP servers and applies them to its configuration
- The switch will pass along what it learned to others

Transparent

- The switch does not participate in the VTP domain
- It does not synchronize its VLAN configuration with other switches
- It can create, modify, and delete VLANs locally, but these changes are not propagated to other switches in the VTP domain

Off

- Disable VTP on the switch



VTP Considerations

VTP messages are only for automatically synchronizing the creation and deletion of VLANs across switches

- They have no impact on VLAN traffic across trunk links, or VLAN pruning

Switches that are transparent, that belong to other VTP domains, or belong to no domain at all can still form trunk links

Nodes on their VLANs will still communicate with other nodes on the same VLAN on other switches



Configuring VTP Switch Mode Example

```
Switch(config)# vtp domain company.com
```

```
Switch(config)# vtp mode server
```

```
Switch(config)# vtp mode client
```

```
Switch(config)# vtp mode transparent
```

```
Switch(config)# vtp mode off
```

```
Switch(config)# end
```

```
Switch# copy running-config startup-config
```



VTP Commands

Command	Description
<code>vtp domain <domain-name></code> <code>vtp domain company.com</code>	Set the VTP domain name
<code>vtp mode {server client transparent off}</code> <code>vtp mode transparent</code>	• Set the switch's VTP role • Default is server
<code>vtp password <password></code> <code>vtp password pass1234</code>	• Configure a password for VTP messages in a VTP domain • All switches in a VTP domain must have the same password

3.8 VLAN Routing

CCNA 200-301 v1.1

Module 3

Why Route Between VLANs?

InterVLAN Connectivity

How to Route Between VLANs

Router On A Stick

VLAN Interface

Best Way to Route Between VLANs

VLAN Interface Commands

Why Route Between VLANs?

By default, VLANs cannot communicate with each other

- The switch cannot connect them

Often you will want devices on different VLANs to communicate with each other

- You have placed servers or other popular destinations together in a single VLAN
- As administrator, you want to be able to remotely manage devices in other VLANs



InterVLAN Connectivity

Properly configured VLANs each have their own subnet ID

All nodes on that VLAN should belong to that subnet

A router / Layer 3 switch can route between them



How to Route Between VLANs

You must use a router to move traffic between VLANs

Well-designed VLANs include these characteristics:

- Each VLAN has its own subnet ID
- All devices in a VLAN belong to the same subnet



How to Route Between VLANs (cont'd)

VLAN routing can be accomplished in three ways

The router has physical connections to each VLAN

A router interface is configured as a trunk port, then divided into sub-interfaces

- One sub-interface for each VLAN
- Best practice is to name the sub-interface after the VLAN ID (e.g. fa0/0.20 - for VLAN 20)

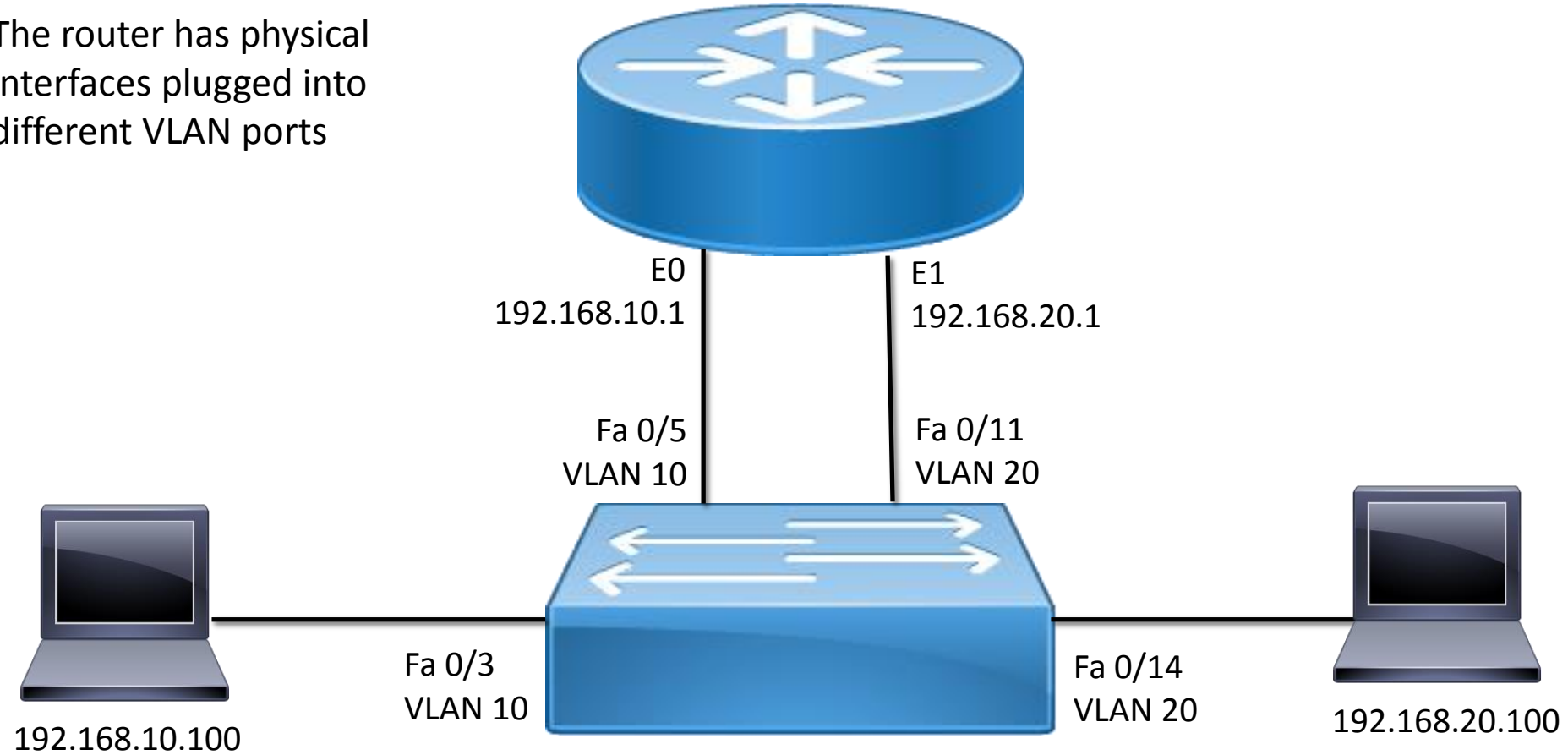
The router is actually a software process inside a multilayer switch

- You need to create a VLAN interface for each VLAN that needs to be routed
- Each VLAN interface must be configured with an IP address, as the default gateway for that VLAN
- All devices on the VLAN must be configured to use that VLAN interface as their default gateway



The Worst Way to Route Between VLANs

The router has physical interfaces plugged into different VLAN ports



Router On A Stick

Some routers allow a port to be configured as a VLAN trunk link

- This configuration is known as “router on a stick”
- On Cisco routers, the port must be Fast Ethernet speed or higher

The physical interface is “divided” into multiple logical sub-interfaces

- The physical interface should not be given an IP address or VLAN ID
- The sub-interfaces are configured with the appropriate IP address and VLAN ID
- Each sub-interface becomes the default gateway for that VLAN
- It is common to name the sub-interface after its associated VLAN ID



Router On A Stick

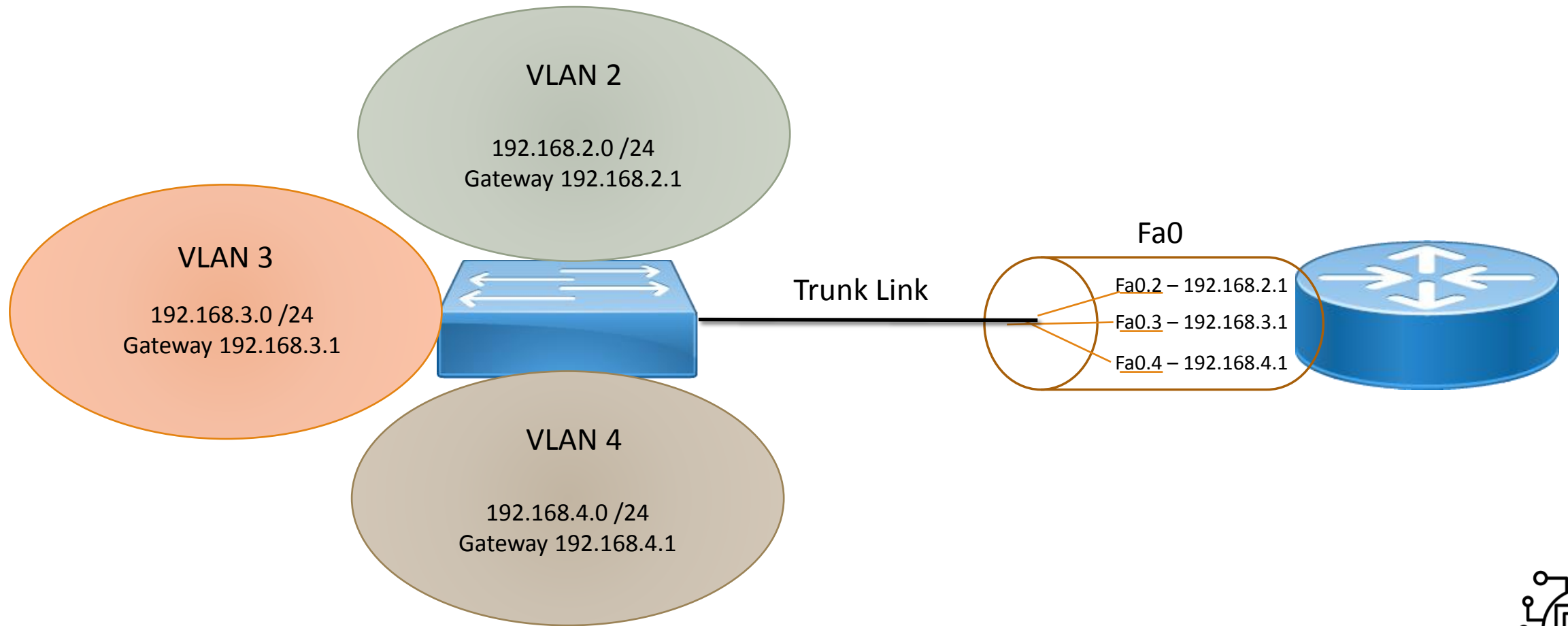
The switch sends tagged traffic down to the trunk link to the router

The router reads the destination IP address, then:

1. Routes the frame to the correct sub-interface
2. Overwrites the Layer 2 header to include the correct destination VLAN
3. The frame is sent back to the switch on the same trunk link
4. The switch forwards the frame out the correct port on the new VLAN



Router on a Stick Example



VLAN Interface



VLAN Interface

AKA Switch Virtual Interface (SVI)

A logical Layer 3 interface configured on a Layer 3 switch

- Typically a Distribution Layer switch

Acts as the default gateway for devices on a VLAN

- The corresponding VLAN needs to exist for you to create a VLAN interface

Should be reachable by all devices in that VLAN

- Can be accessed by devices across any number of trunk links

Once created, will remain up so long as:

- At least one active port belongs to that VLAN
- OR - an active trunk link exists that carries traffic for that VLAN

A switch can have an SVI for every VLAN

Each switch can only have one SVI per VLAN

You can still have multiple SVIs in the same VLAN (one per switch)



Configuring a VLAN Interface Example

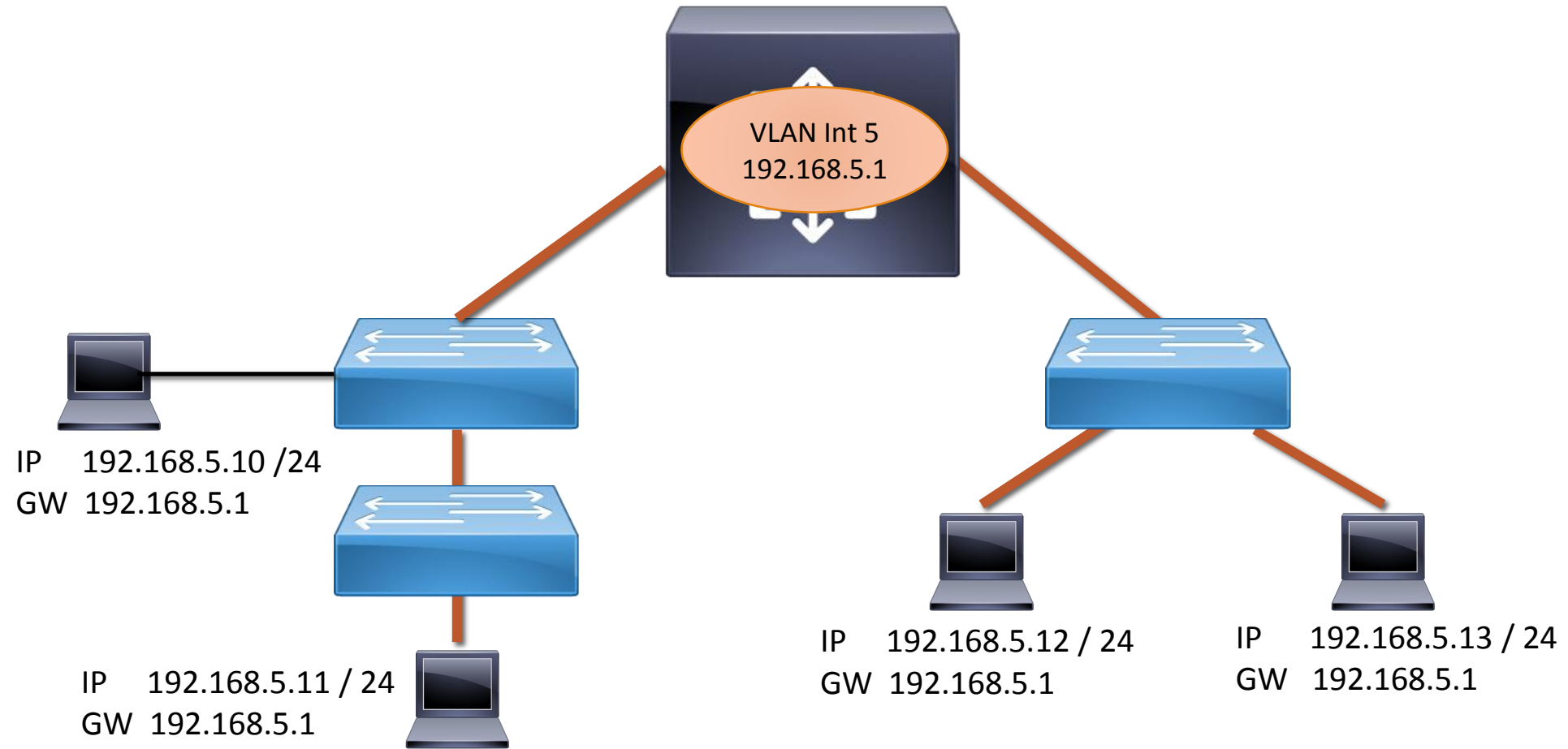
```
Switch(config)# interface vlan 5
```

```
Switch(config-if)# ip address 192.168.5.1 255.255.255.0
```

```
Switch(config-if)# no shut
```



VLAN Interface Example



Best Way to Route Between VLANs

Use VLAN interfaces (SVIs) as default gateways for their respective VLAN

Ensure that:

- The VLAN already exists
- There is at least one port assigned to the VLAN with a live device connected to it
 - To keep it in an up state
- Alternative to the live device, you can have a live trunk link that carries the VLAN
 - No other member port is required



Best Way to Route Between VLANs (cont'd)

Configure a VLAN interface as the VLAN's default gateway

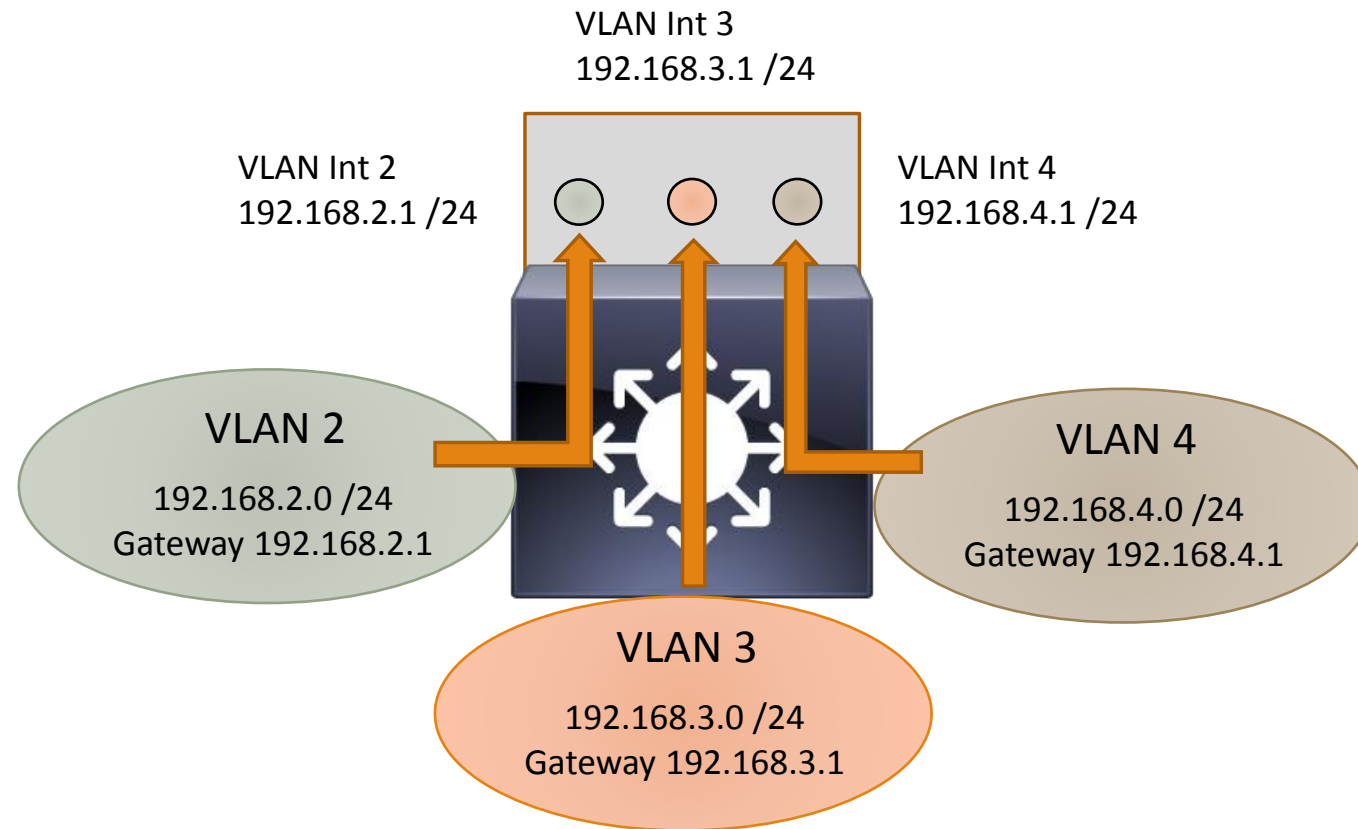
- The VLAN interface name also determines its VLAN membership (e.g. "interface vlan 5" automatically belongs to VLAN 5)
- The default gateway IP address for that VLAN's subnet
- Ensure that all nodes in the VLAN can ping their VLAN interface default gateway (even across switch trunk links)

Note:

- VLAN interfaces route between VLANs on the backplane of the switch
- VLAN interfaces can therefore only route between Ethernet segments
- VLAN interfaces stay "up" so long as at least one VLAN member port is active
- A single trunk link can also keep all VLAN interfaces up



VLAN Routing Example



Configuring SVI VLAN Routing Example

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 5
```

```
Switch(config)# interface gigabitEthernet 0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 5
```

```
Switch(config-if)# int vlan 5
```

```
Switch(config-if)# ip address 192.168.5.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

```
Switch# copy running-config startup-config
```



VLAN Interface Commands

Command	Description
<code>ip routing</code>	(If necessary) enable IP routing on the switch
<code>interface vlan [vlan ID]</code> <code>interface vlan 10</code>	Enter VLAN interface mode
<code>ip address [address] [subnet mask]</code> <code>ip address 192.168.10.1 255.255.255.0</code>	Assign an IP address to the VLAN interface
<code>no shutdown</code>	Administratively turn the port on
<code>show ip interface brief</code>	Check the VLAN int status
<code>show ip route</code>	Show the routing table Confirm that the switch is ready to route traffic between VLANs

3.9 Switchport Configuration

CCNA 200-301 v1.1

Module 3

Common Port Configurations

Switchport Modes

Phone and PC VLANs

Phone and PC VLAN Commands

Port Mirroring

Jumbo Frames

Straight Through and Crossover Cables

Flow Control

IGMP Snooping

Power over Ethernet

PoE Commands

Port Security

MAC Aging

MAC Aging Commands

Port Security with Voice VLANs

Data and Voice Port Security Commands

Multilayer Switch Routed Ports

Routed Port Commands

Common Port Configurations

Speed

- 10mb / 100mb / 1000mb / 10g, etc.
- Auto - Device and switchport will negotiate the fastest speed both support

Duplex

- Full
 - Both sides can transmit simultaneously
 - Requires two separate conductors/wire pairs (Tx and Rx)
- Half
 - Devices must take turns transmitting
- Auto – Device and switchport will attempt to negotiate full duplex if possible

Mode

- Access – port is for end devices
- Trunk – port connects to another switch to extend VLAN



Switchport Modes

Each port on a switch can be configured for one of two modes:

- Access
- Trunk

You can change the mode on a port as necessary

Access Port	Trunk Port
Connects to an end device	Connects to another switch
Belongs to one VLAN	Used by all VLANs
Frames transmitted out the port are not tagged	Frames transmitted out the port are “tagged” with the VLAN they belong to
You can use an uplink (regular crossover cable) to plug a hub into the port to add more devices to the port	After connecting the switches with a crossover cable, you configure the ports on both sides for trunking



Phone and PC VLANs



Phone and PC VLANs

1. Some switches allow you to create a second “phone” VLAN on the same port
2. You then plug the phone into the switchport, and the PC into the phone
3. After configuring the switchport for the PC VLAN, you can add the Voice VLAN on the same interface:

```
Switch(config-if)# switchport voice vlan [voice-vlan-id]
```

5. You should also configure QoS to give voice traffic priority:

```
Switch(config-if)# mls qos trust cos
```

6. Set the switchport to portfast so there is no delay when the user wants to use the phone

```
Switch(config-if)# spanning-tree portfast
```



Phone and PC VLAN Configuration Example

```
Switch# configure terminal
Switch(config)# interface gigabitEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# switchport voice vlan 20
Switch(config-if)# mls qos trust cos
Switch(config-if)# spanning-tree portfast
Switch(config-if)# end

Switch# copy running-config startup-config

Switch# show running-config interface gigabitEthernet 0/1
```



Phone and PC VLAN Commands

Configuration	Description
<pre>switchport access vlan [data-vlan-id] switchport access vlan 5</pre>	Assign the data VLAN to the switchport that will have the phone and PC
<pre>switchport voice vlan [voice-vlan-id] switchport voice vlan 7</pre>	Add the voice VLAN to the same switchport
<pre>mls qos trust cos</pre>	Configure QoS to give voice traffic priority
<pre>spanning-tree portfast</pre>	Set the switchport to portfast so there is no delay when the user wants to use the phone



Port Mirroring



Switched Port Analyzer (SPAN)

AKA port spanning, port mirroring

All traffic on this port is copied to another port

Admin puts a protocol analyzer or sniffer on the mirrored port to monitor traffic

You can have multiple SPAN sessions on a switch, identified by a session number (typically 1-66, depending on the switch model)

```
Switch(config)# monitor session [session_number] source interface [type] [port] [direction]
```

```
Switch(config)# monitor session 1 source interface FastEthernet0/1 both
```

For information on configuring SPAN see:

<https://www.cisco.com/c/en/us/support/docs/switches/catalyst-6500-series-switches/10570-41.html>



Jumbo Frames



Jumbo Frames

Configure specific switchports to allow Ethernet frame size up to 9000 bytes

- Ensure that the end devices (servers, storage controllers, etc.)

Beneficial in storage networks, high-performance computing, and certain data center environments

- Useful for iSCSI or other links where you are trying to increase performance
- Reduces the number of frames needed for large data transfers

```
Switch(config)# system mtu jumbo 9000
```

```
Switch(config-if)# mtu 9000
```

For information on configuring jumbo frames see:

<https://www.cisco.com/c/en/us/support/docs/switches/catalyst-6000-series-switches/24048-148.html>



Straight Through and Crossover Cables



Straight Through and Crossover Cables

The transmit wires of one device should connect to the receive wires of the other device

- You cannot communicate if transmit is connect to transmit, or receive is connected to receive

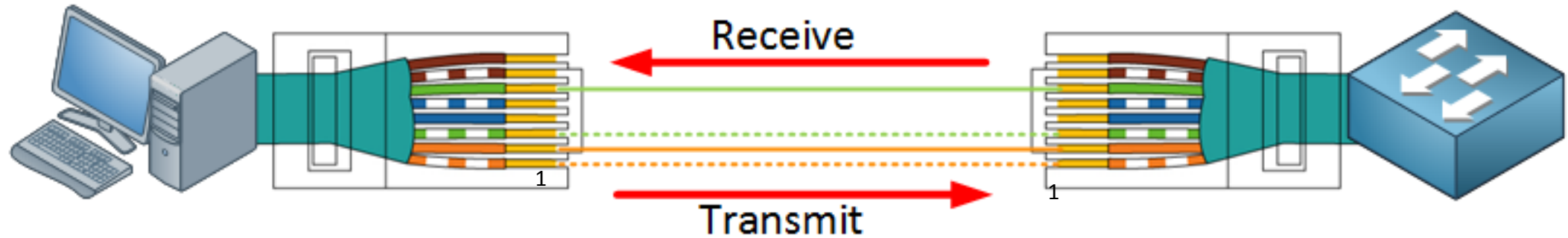
On modern switches, the ports can detect if transmit and receive signals are connected to the wrong pins

- Will automatically swap Tx and Rx ports if the wrong cable was used
- The ports cannot correct other types of miswiring

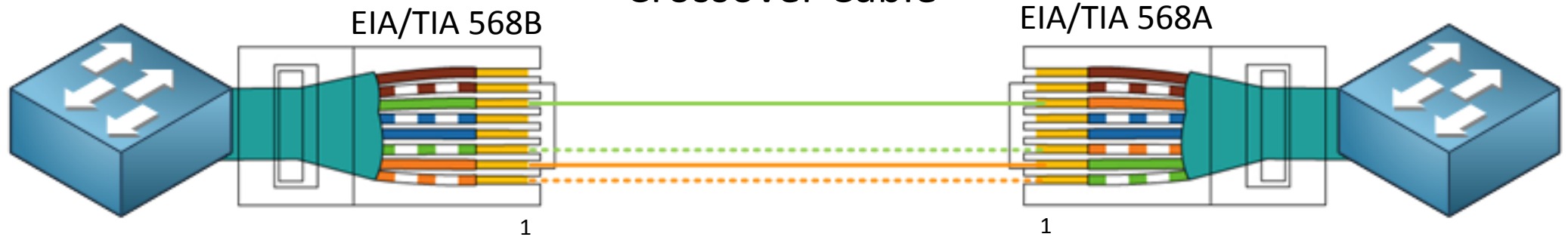


Ethernet Straight-through and Crossover Cable Wiring

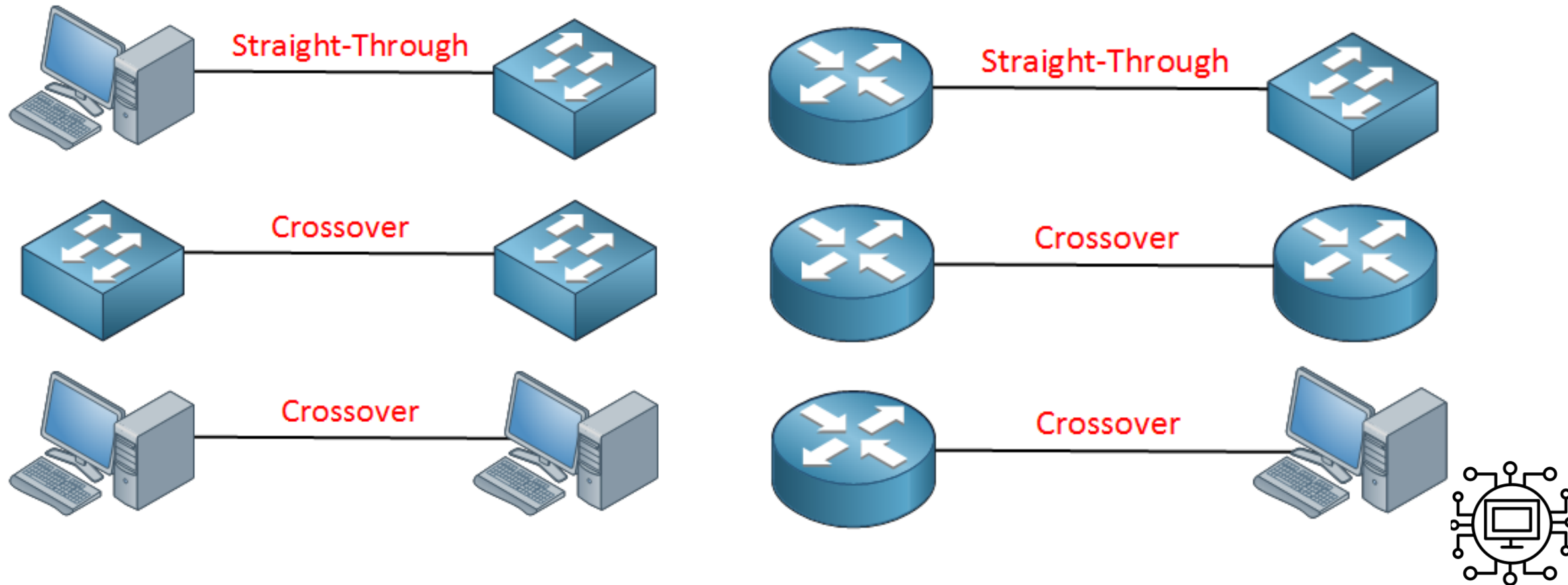
Straight-through Cable EIA/TIA 568B



Crossover Cable



Ethernet Straight-Through / Crossover Cable Selection



Flow Control



Flow Control

A predictive congestion management mechanism

Used by a switch to prevent uncontrolled packet drops

- The switch PREDICTS that based on the current traffic flow, it will run out of receive buffer space in the next few frames
- It sends a PAUSE frame (request) to the sending device
- The sending device shuts up for a few milliseconds
- This is a complete halt of all traffic flow, regardless of traffic priority

This mechanism is no longer preferred

- It is better to implement and enforce true Quality of Service (QoS)
- Even better to monitor bandwidth utilization trends and increase bandwidth / offload traffic to other segments

For more information on configuring switch QoS see:

<https://www.cisco.com/c/en/us/support/docs/switches/catalyst-3750-series-switches/91862-cat3750-qos-config.html>



IGMP Snooping



IGMP Snooping

Distinguishes between different multicast sessions

- Tracks which ports have subscribed to which multicast groups

You configure the switch to watch for IGMP messages from clients

- It's often enabled by default on Cisco switches
- The switch will only forward multicasting for a particular group out the relevant ports

A multicast group is identified by the IP address it uses

- IP range is 224.0.0.0 to 239.255.255.255
- Example: You could have two multicast groups, 224.1.1.1, and 224.1.1.2
- Host A only wants traffic for 224.1.1.1
- Host B only wants traffic for 224.1.1.2
- With IGMP snooping, the switch will only forward multicasting for each group out the appropriate port

Forward multicast frames out specific ports, rather than flooding

Configured at the switch level



IGMP Snooping Example

```
Switch# configure terminal
```

```
Switch(config)# ip igmp snooping
```

For more information on configuring IGMP snooping see:

<https://www.cisco.com/c/en/us/support/docs/smb/switches/cisco-350x-series-stackable-managed-switches/smb5814-configure-igmp-switch-cli.html>



Power over Ethernet (PoE)



Power over Ethernet (PoE)

Some switches can provide PoE on their ports

- Provides power to WAPs, IP phones, Cameras, networked industrial controls, alarms, etc.
- Power is DC

Useful when there is no convenient power source near the end device

If the switch does not support PoE, you can also insert a separate power injector between the switch and the end device



PoE Considerations

Be mindful of the switch's overall power budget

- You do not want to overload the switch past its capability
- That would cause some ports to not provide power, or require you to prioritize certain ports.

Only twisted pair cable carries PoE

- Use CAT 6a or higher for distances over 50 meters
- Don't be surprised at 25% power loss by the time PoE reaches the end device



PoE Considerations (cont'd)

The PoE port must be directly connected to the PoE device

- Do not insert another switch or hub between the PoE switchport and its end device
- The intermediate device cannot “relay” power

Most PoE switches can autosense if the end device requires PoE

- Will automatically turn the power on or off

If your switch does not support PoE, then use a power injector

- Place the injector as close as possible to the end device

The average life span of a PoE switch is 3 – 5 years



PoE Standards

PoE 15.4 watts (802.3af)

Uses 2 pairs to carry power



VoIP



Wi-Fi



PoE Standards

PoE 15.4 watts (802.3af)

Uses 2 pairs to carry power



VoIP



Wi-Fi

PoE+ 30 watts 802.3at

Uses 2 pairs to carry power

Cisco Universal Power over Ethernet
(UPOE) enhances PoE+ at 60 watts



Pan/Tilt/Zoom Cameras



Video IP Phones



Alarm Systems



PoE Standards

PoE 15.4 watts (802.3af)

Uses 2 pairs to carry power



VoIP



Wi-Fi

PoE+ 30 watts 802.3at

Uses 2 pairs to carry power

Cisco Universal Power over Ethernet (UPOE) enhances PoE+ at 60 watts



Pan/Tilt/Zoom Cameras



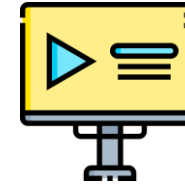
Video IP Phones



Alarm Systems

PoE++ 90 watts (802.3bt)

Aka 4PPoE - uses all 4 pairs



Digital Signage



Advanced
Video IP
Phones



Networked
Industrial
Control Systems



Configure PoE Globally Example

```
Switch# configure terminal
```

```
# Set the total switch power budget
```

```
Switch(config)# power inline power-limit [watts]
```

```
Switch(config)# power inline power-limit 400
```

```
# Set the default PoE priority level for all switchports
```

```
Switch(config)# power inline port priority [ critical | high | low ]
```

```
Switch(config)# power inline port priority high
```



Configure PoE Per Interface

```
Switch# configure terminal
```

```
Switch(config)# interface [interface-id]
```

```
# Turn PoE on (if necessary)
```

```
Switch(config-if)# power inline auto
```

```
# Turn PoE off (that interface)
```

```
Switch(config-if)# power inline never
```

```
Switch(config-if)# power inline static max [max-milliwatts]
```

```
# If the switch supports power priority, set critical, high, or low for  
the device on that port
```

```
Switch(config-if)# power inline priority [level]
```



PoE Commands

Configuration	Description
<code>power inline auto</code>	Automatically detect and supply power to connected devices that are PoE-capable (e.g. IP phones, WAPs, security cameras)
<code>power inline never</code>	Disable PoE on a specific interface
<code>power inline static max [max-milliwatts]</code> <code>power inline static max 30000</code>	Configure a maximum amount of power (in milliwatts) that a specific port can supply to connected devices
<code>power inline priority [critical high low]</code> <code>power inline priority critical</code>	Set priority for that port (connected device) in case the PoE budget becomes strained and the switch has to make decisions about which devices to power



Port Security



Port Security

Port security can only be configured on access ports

- `Switch(config-if)# switchport port-security`

MAC address limiting

- Restrict how many MAC addresses are allowed on that port
- `Switch(config-if)# switchport port-security maximum [number]`

Static MAC

- Manually associate a MAC (and only that MAC) on a port
- `Switch(config-if)# switchport port-security mac-address [mac-address]`

Sticky MAC

- First dynamically learned MAC address becomes the authorized MAC for that port
- Saved in running-config / retained even after a reboot
- `switchport port-security mac-address sticky`



Port Security Violation Modes

You can set the Violation Mode to determine what the switch will do

- `Switch(config-if)# switchport port-security violation [protect | restrict | shutdown]`

Shutdown

- The default mode
- The port is immediately put into the error-disabled state, effectively shutting it down
- Port must be manually re-enabled or configured to automatically recover after a certain period

Protect

- Discards frames with unauthorized MAC addresses
- Does not generate a security notification or disable the port

Restrict

- Discards frames with unauthorized MAC addresses
- Increments the security violation counter
- Generates a log message or SNMP trap
- Does not disable the port



Port Security Commands

Configuration	Description
<code>switchport port-security</code>	Enable port security on a specific interface
<code>switchport port-security maximum [number]</code> <code>switchport port-security maximum 2</code>	Set the maximum number of allowed MAC addresses on a port
<code>switchport port-security mac-address [mac-address]</code> <code>switchport port-security mac-address 0001.2345.6789</code>	Manually specify the MAC address that is allowed in the interface
<code>switchport port-security mac-address sticky</code>	• Add the first MAC address heard on the port as the authorized MAC • Save in running config
<code>switchport port-security violation [protect restrict shutdown]</code> <code>switchport port-security violation protect</code>	• Specify the violation response • Default is shutdown



MAC Aging

Ensure the switch does not keep stale (or even malicious) entries in the MAC table

MAC addresses are removed after a period of inactivity

Can be set from 0 (never remove) to 1,000,000 seconds

Default is 300 seconds (5 minutes)



Configuring MAC Aging

Configure MAC aging time globally for all VLANs

```
Switch(config)# mac address-table aging-time [time]
```

```
Switch(config)# mac address-table aging-time 600
```

Configure MAC aging time for a specific VLAN

```
Switch(config)# mac address-table aging-time [time in seconds]  
vlan [vlan-id]
```

```
Switch(config)# mac address-table aging-time 300 vlan 10
```

Verify MAC aging time

```
Switch# show mac address-table aging-time
```



MAC Aging Commands

Configuration	Description
<code>mac address-table aging-time [time]</code> <code>mac address-table aging-time 600</code>	Configure MAC aging time globally for all VLANs
<code>mac address-table aging-time [time in seconds] vlan [vlan-id]</code> <code>mac address-table aging-time 300 vlan 10</code>	Configure MAC aging time for a specific VLAN
<code>show mac address-table aging-time</code>	Verify MAC aging time



Port Security with Voice VLANs

Switchport security can be set to allow voice and data VLANs to coexist on a switchport

```
Switch(config-if)# switchport voice vlan [voice-vlan-id]
```

1. Configure the switchport for both the access (data) vlan and the voice vlan

```
Switch(config-if)# switchport access vlan 5
```

```
Switch(config-if)# switchport voice vlan 7
```

2. Configure port security normally

- BE SURE to allow 2 MAC addresses on the port

```
Switch(config-if)# switchport port-security maximum 2
```



Configuring Port Security for Voice and Data Example

```
Switch# configure terminal
Switch(config)# interface g0/10
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 5
Switch(config-if)# switchport voice vlan 7
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security violation protect
```



Data and Voice Port Security Commands

Configuration	Description
<pre>switchport voice vlan [vlan ID] switchport voice vlan 7</pre>	Configure the switch access port with the voice VLAN ID
<pre>switchport port-security maximum [number] switchport port-security maximum 2</pre>	Limit the number of allowed MAC addresses on the switchport
<pre>switchport port-security violation [protect restrict shutdown] switchport port-security violation protect</pre>	Set the violation response on the switchport



Multilayer Switch Routed Ports



Multilayer Switch Routed Ports

Some switch models allow you to toggle a port from Layer 2 (switchport) to Layer 3 (routed port)

- Catalyst 3560, 3750, 3850, 4500, and 6500 series

Routed ports cannot be part of a VLAN

- A routed port is treated as a standalone Ethernet router interface
- If you wish to route VLAN traffic, you will need a VLAN interface
- The switch will disallow two interfaces (be they routed port or SVI) from belonging to the same subnet
- A Layer 3 switch will route between all subnets it has an interface to, be they SVIs or Routed ports

You can physically connect a network segment/subnet to the routed port to be the default gateway for nodes on that segment

- You must use a simple uplink to make the connection
- The routed port has no concept of what a VLAN is



Routed Port Commands

Command	Description
<code>interface [type][#]</code> <code>interface g0/1</code>	Enter interface configuration mode
<code>no switchport</code>	Convert a switchport from Layer 2 to Layer 3
<code>ip address [address] [subnet mask]</code> <code>ip address 10.1.1.1 255.255.255.0</code>	Assign an IP address and subnet mask
<code>no shutdown</code>	Turn the port on
<code>show ip route</code>	• View the route table • SVIs and Routed ports will show as “directly connected”

3.10 EtherChannel

CCNA 200-301 v1.1

Module 3

What is EtherChannel?

Static EtherChannel Configuration

Automatic PAgp EtherChannel Configuration

Automatic LACP EtherChannel Configuration

EtherChannel Commands

EtherChannel Show Commands

What is EtherChannel?

AKA link aggregation, channel group

Cisco's implementation of a Link Aggregation Group (LAG - IEEE 802.1AX)

The bundling of multiple physical Ethernet links into a single logical link

Allows switches with multiple links to negotiate and bundle the links together

The aggregate link provides increased bandwidth, redundancy, and load balancing

- If one member link goes down, the EtherChannel continues to function

It is treated as a single link by the network devices

Also treated as a single link by Spanning Tree

Note: The term "EtherChannel" refers to aggregating multiple Ethernet links into a single logical link. The link has underlying individual physical switchports that become "EtherChannel member ports"



EtherChannel Uses

Once you create the EtherChannel, you can decide how to use it

- Ensure that you configure both sides of the EtherChannel link for the use case

Access link to an end device

- Some computer/storage controller Ethernet ports support EtherChannel between the end device and its access switch
- Be careful to ensure that all ports on both sides are in the same VLAN

Simple uplink to another switch

- You can leave all ports on the downstream switch (including the EtherChannel ports) in the default VLAN (VLAN 1)

Layer 2 trunk link between switches

- Turn the EtherChannel into a trunk link

Layer 3 routed port link

- Turn the EtherChannel into a link between a router port and another device (end device or switch)



EtherChannel Configuration

You can configure EtherChannel several ways:

Manually

- Static EtherChannel doesn't use a protocol for negotiation
- The EtherChannel is manually configured
- There is no attempt to negotiate with the other side
- `Switch(config-if)# channel-group 1 mode on`

Automatic negotiation

- PAgp (Port Aggregation Protocol)
 - A Cisco proprietary protocol for negotiating the creation of EtherChannel links
 - `Switch(config-if)# channel-group 1 mode desirable`
- LACP (Link Aggregation Control Protocol)
 - The IEEE 802.3ad standard for negotiating EtherChannel links
 - `Switch(config-if)# channel-group 1 mode active`



Static EtherChannel Configuration



Static EtherChannel Access Link Example

Enter interface configuration mode for interfaces you want to add to the EtherChannel group

```
Switch(config)# interface range gigabitEthernet0/1-2
```

Assign the physical interfaces to an EtherChannel group;

```
Switch(config-if-range)# channel-group 1 mode on
```



Static EtherChannel Trunk Link Example

Enter interface configuration mode for interfaces you want to add to the EtherChannel group

```
Switch(config)# interface range gigabitEthernet0/1-2
```

Assign the physical interfaces to an EtherChannel group;

```
Switch(config-if-range)# channel-group 1 mode on
```

(For older switches) Specify the EtherChannel encapsulation mode if necessary

```
Switch(config-if-range)# switchport trunk encapsulation dot1q
```

Change the access port to a trunk port

```
Switch(config-if-range)# switchport mode trunk
```



Static EtherChannel Routed Port Example

```
# Configure the physical interfaces for EtherChannel
Switch(config)# interface range g0/1-2
Switch(config-if)# channel-group 1 mode on

# Convert the port-channel interface to a routed port
Switch(config-if)# interface port-channel 1
Switch(config-if)# no switchport

# Assign an IP address to the routed port-channel interface
Switch(config-if)# ip address 192.168.1.1 255.255.255.0

# Check the port-channel interface configuration
Switch(config-if)# end

Switch# show running-config interface port-channel 1

# Display a summary of interfaces, including port-channel 1
Switch# show ip interface brief
```



Automatic PAgp EtherChannel Configuration



Dynamic EtherChannel with PAgp Example

Enter interface configuration mode for interfaces you want to add to the EtherChannel group

```
Switch(config)# interface range gigabitEthernet0/1-2
```

Use PAgp for negotiation, with the mode set to desirable (actively tries to form an EtherChannel) or auto (passively waits to form an EtherChannel)

```
Switch(config-if-range)# channel-group 1 mode desirable
```

(Alternatively) Use auto mode if you want the interface to wait passively for the other end to initiate the EtherChannel

```
Switch(config-if-range)# channel-group 1 mode auto
```



Dynamic EtherChannel with PAgp Trunk Link Example

Enter interface range configuration mode for the physical interfaces

```
Switch(config)# interface range gigabitEthernet0/1-2
```

Use PAgp to actively attempt to create the EtherChannel (channel-group)

```
Switch(config-if-range) # channel-group 1 mode desirable
```

Configure the physical interfaces (and thus the channel-group) start trunking

```
Switch(config-if-range) # switchport mode trunk
```



Dynamic EtherChannel with PAgp Routed Port Example

Create the channel-group and then return to global configuration mode

```
Switch(config)# interface range gigabitEthernet0/1-2
```

```
Switch(config-if-range)# channel-group 1 mode desirable
```

```
Switch(config-if-range)# exit
```

Create the logical port-channel interface

```
Switch(config)# interface port-channel 1
```

Turn the port-channel interface into a routed port

```
Switch(config-if)# no switchport
```

Assign an IP address to the routed port-channel interface

```
Switch(config-if)# ip address 192.168.1.1 255.255.255.0
```



Automatic LACP EtherChannel Configuration



Dynamic EtherChannel with LACP Example

Enter interface configuration mode for interfaces you want to add to the EtherChannel group

```
Switch(config)# interface range gigabitEthernet0/1-2
```

Use LACP for negotiation, with the mode set to active (actively tries to form an EtherChannel) or passive (waits passively to form an EtherChannel)

```
Switch(config-if-range)# channel-group 1 mode active
```

(Alternative) Use passive mode if you want the interface to wait passively for the other end to initiate the EtherChannel.

```
Switch(config-if-range)# channel-group 1 mode passive
```



Dynamic EtherChannel with LACP Trunk Example

Create the EtherChannel using LACP

```
Switch(config)# interface range gigabitEthernet0/1-2
```

```
Switch(config-if-range)# channel-group 1 mode active
```

```
Switch(config-if-range)# exit
```

(For older switches) Specify the EtherChannel encapsulation mode if necessary

```
Switch(config-if-range)# switchport trunk encapsulation dot1q
```

Configure the physical interfaces (and thus the channel-group) start trunking

```
Switch(config-if-range)# switchport mode trunk
```



Dynamic EtherChannel with LACP Routed Port Example

Create the EtherChannel using LACP

```
Switch(config)# interface range gigabitEthernet0/1-2
```

```
Switch(config-if-range)# channel-group 1 mode active
```

```
Switch(config-if-range)# exit
```

Create the logical port-channel interface

```
Switch(config)# interface port-channel 1
```

Turn the port-channel interface into a routed port

```
Switch(config-if)# no switchport
```

Assign an IP address to the routed port-channel interface

```
Switch(config-if)# ip address 192.168.1.1 255.255.255.0
```



EtherChannel Commands

Command	Description
<code>channel-group [number] mode on</code> <code>channel-group 1 mode on</code>	Configure static EtherChannel
<code>channel-group [number] mode desirable</code> <code>channel-group 1 mode desirable</code>	Configure physical ports to actively negotiate creating an EtherChannel using PAgp
<code>channel-group [number] mode auto</code> <code>channel-group 1 mode auto</code>	Configure physical ports to not actively negotiate, but be willing to form an EtherChannel using Pagp if the other side is set to <code>mode desirable</code>
<code>channel-group [number] mode active</code> <code>channel-group 1 mode active</code>	Configure physical ports to actively negotiate creating an EtherChannel using LACP
<code>channel-group [number] mode passive</code> <code>channel-group 1 mode passive</code>	Configure physical ports to not actively negotiate, but be willing to form an EtherChannel using LACP if the other side is set to <code>mode active</code>



EtherChannel Show Commands

Command	Description
<code>show etherchannel</code> <code>show etherchannel summary</code>	Check the group state The various channel groups and member ports
<code>show lacp neighbor</code> <code>show pagp neighbor</code>	Information about the other side of the link
<code>show etherchannel port-channel</code>	Check port-channel status
<code>show interfaces port-channel</code>	Display information on all port-channels

Review

3.11 Review

CCNA 200-301 v1.1

Module 3

Review

Ethernet is the most commonly used Layer 2 wired LAN protocol

Ethernet uses source and destination MAC addresses to pass frames from one host to another on the same segment

Ethernet has a minimum length of 64 bytes, and a maximum length (MTU) of 1500 bytes

Network contention occurs when multiple nodes try to use the same shared network segment at the same time

Ethernet uses CSMA/CD to manage the impact of collisions

A collision domain is a network segment such as a hub, half-duplex switchport, or coax bus where collisions can occur

Switches in full duplex mode micro-segment a network into multiple collision domains, one for each switchport



Review (cont'd)

A broadcast domain is a network segment in which a Layer 2 broadcast will be “heard” by all hosts

- Can be a physical segment or a VLAN

Routers block broadcasts

Each router interface has its own broadcast domain

Routers create broadcast domains

Switches create collision domains

Every broadcast domain will have at least one collision domain



Review (cont'd)

A switch is a Layer 2 device that makes forwarding decisions based on Layer 2 MAC addresses

A switch listens to frames coming into its ports. The source MAC address of the frame is placed into the switch's MAC address table

The switch's MAC address table maps MAC addresses to switch ports

When the switch receives an incoming frame and must make a decision on which switchport to forward the frame out of, it refers to its MAC table to determine the correct port

MAC learning is the process by which a switch populates its MAC address table

MAC aging is the process by which entries in the MAC address table are removed after a certain period of inactivity



Review (cont'd)

Frame switching is the process of forwarding incoming frames to the correct destination Port as listed in the MAC table

A switch will flood broadcasts, multicasts and unknown unicasts out all ports including down trunk links, except for the port the frame was received on



Review (cont'd)

A switch can be configured to run in one of three modes:

Store and forward will buffer an entire incoming frame and run a CRC check to make sure the frame is not damaged before forwarding it to its destination

Fragment-free will validate the first 64 bytes of an incoming frame and then forward it to its destination

Cut through does not check the frame at all, instead trading accuracy for performance



Review (cont'd)

A switching loop occurs when there are redundant links between switches

The spanning tree protocol (STP) is used to eliminate ethernet switching loops

The spanning tree process starts by the switches comparing their Bridge IDs to elect a root Bridge

The Root Bridge is the switch that has the lowest priority value

- In case of a tie, the switch with the lowest MAC address will win the election

You can manually configure the priority value of switches to influence whether or not they are likely to win the root Bridge election



Review (cont'd)

In STP, switch ports move through several states before they forward user traffic

In the **Blocking State** the port listens for Bridge Protocol Data Units (BPDUs) but does not forward any traffic

In the **Listening State** the port is preparing to forward data but does not yet

- It is listening to see if BPDUs are coming in on multiple ports, an indication of a switching loop

In the **Learning State** the port begins to learn what MAC addresses are on it

- It is still not yet forwarding any traffic

In the **Forwarding State** the port is finally forwarding user traffic, as well as learning new MAC addresses and listening for BPDUs

The **Disabled State** is not actually an STP State

- If the administrator shuts a port down, that port will not participate in STP at all



Review (cont'd)

A port can play one of three roles in STP

The **Root Port** is the port on a non-root switch that has the lowest overall cost path back to the root Bridge

Each non-root switch has one Root Port

A **Designated Port** is one that is forwarding traffic

- It has the lowest cost path for that particular network segment back to the Root Bridge
- Every network segment will have one Designated Port

A **Non-designated Port** is one that has been placed in a blocking state by the switch

- To break switching loops
- Non-designated Ports do not forward traffic
- They are still considered active and are listening for STP updates



Review (cont'd)

A Root Bridge does not have a root port

- All of its ports are considered Designated
- A Root Bridge will not place any of its own ports in a blocking state

Non-root switches put their own redundant ports in a blocking state to interrupt switching loops

- If the switch receives Root BPDUs on more than one port it knows it is participating in a switching loop

The switch will place the least important redundant port in a blocking state

- Switches determine which Port is the least important by looking at the total cost of the path from that port back to the Root Bridge
- The Redundant port with the most expensive path is put in a blocking state
- If there are multiple paths with the same cost back to the root Bridge, then the switch will block the port that has the highest interface number



Review (cont'd)

Rapid PVST+ is a Cisco-proprietary enhancement to STP that provides much faster convergence, as well as a spanning-tree instance for every VLAN

The three port states in Rapid PVST+ are Discarding (a combination of blocking and listening), Learning, and Forwarding

In Rapid PVST+, a port can play one of five roles

- The **Root Port** is the port on a non-root bridge with the best overall path cost back to the root
- The **Designated Port** is allowed to forward frames on a given link. It has the lowest path cost to the root bridge for that segment
- An **Alternate Port** is a standby backup port that can take over in case the current root port fails
- A **Backup Port** is a redundant connection to a designated port
 - It is typically found in scenarios with redundant links to the same network segment
- A **Disabled Port** is one that has been administratively shut down and is not participating in spanning-tree



Review (cont'd)

A VLAN is a logical grouping of access switchports

- A modern switch supports 1005 VLANs
- Some switches support 4096 total VLANs

Devices plugged into a switchport have no awareness of being in a VLAN

VLANs can be extended across switches via trunk links

A VLAN is also a broadcast domain

Ports in a VLAN can only communicate with ports in the same VLAN

To implement a VLAN properly, it should also be its own subnet

Switches cannot connect VLANs together

A router is required to route traffic between VLANs



Review (cont'd)

Initially, every port on a switch belongs to the default, VLAN 1

Devices that are plugged into a switchport are unaware that they belong to a particular VLAN

You can add more VLANs to the switch, and then assign a switchport to whichever VLAN

Ordinarily, a switchport can only belong to one VLAN at a time

Some switches support having both a Data VLAN and a Phone VLAN on the same port



Review (cont'd)

You can connect two switches together either through a simple uplink or a trunk link

In an uplink, the downstream switch becomes an extension of the one switchport it is plugged into

The downstream switch should not be configured with any VLANs

All devices plugged into the downstream switch will belong to the same VLAN that the upstream switchport belongs to

In a trunk link, the two switches on either side of the link configure their ports to be trunk ports

Trunk ports belong to no particular VLAN

Trunk links carry traffic for all VLANs between the switches

Trunk links make it possible to extend a VLAN across many switches

- An ARP broadcast on a VLAN will be sent down trunk links, from switch to switch, so that it can be received by all nodes in that VLAN



Review (cont'd)

802.1q “dot1q” is the most commonly used VLAN trunking protocol in use today

When a frame is sent down a trunk link, dot1q tags the frame with the VLAN it belongs to

- The receiving switch then knows which VLAN to deliver the frame to

A VLAN tag is ignored by any device (such as a hub or third party switch) that does not support dot1q

If a port ceases to be a trunk port, it will revert to its Native VLAN (by default VLAN 1)

For security purposes, you should change the Native VLAN on switches to something other than VLAN 1

You will need to configure the same native VLAN on all switches so there are no VLAN mismatches

Frames that belong to the Native VLAN will not be tagged, regardless of what the Native VLAN number is



Review (cont'd)

VLAN pruning allows you to disallow certain VLANs from being sent or received from a trunk port

VLAN pruning reduces the amount of unnecessary bandwidth consumed on a link by broadcasts and other messages that Downstream switches do not need to receive

The VLAN trunking protocol (VTP) is a Cisco proprietary protocol used by switches to inform each other of new or removed VLANs

Switches must belong to the same VTP domain in order to share VTP updates

You can optionally set a VTP password to help prevent VTP information from being sent to or received from the wrong switches



Review (cont'd)

A switch can be configured to be in one of several VTP modes

A VTP server is a switch that can create, modify and delete VLANs

A VTP client is a switch that cannot create, modify or delete VLANs

- It learns about VLANs from Servers or other Clients
- It passes along what it has learned to other switches

A VTP server switch can learn about new VLANs from a VTP server or client



Review (cont'd)

VTP switch in transparent mode does not participate in the VTP domain

- You will have to create its VLANs manually
- Transparent mode switches will pass along any VTP messages they receive, without acting upon those messages themselves

VTP messages are only for automatically synchronizing the creation and deletion of VLANs across switches

- They have no impact on VLAN traffic across trunk links, or VLAN pruning
- Switches that are transparent, that belong to other VTP domains, or belong to no domain at all can still form trunk links
- Nodes on their VLANs will still communicate with other nodes on the same VLAN on other switches



Review (cont'd)

If you want nodes on different VLANs to communicate with each other, you will need to route between the VLANs

Historically, a common way to route between VLANs was to create a “router on a stick”

- A router’s FastEthernet port was divided into logical subinterfaces
- Each subinterface would then be assigned to a different VLAN and be the default gateway for that VLAN

The best way to route between VLANs today is to create a logical VLAN interface (aka SVI) on a Layer 3 switch (Distribution Layer switch)

This VLAN interface belongs to the VLAN it is assigned to

- It is given an IP address and subnet mask
- It acts as the default gateway for all nodes on that VLAN

The VLAN interface needs to be reachable by all devices in that VLAN

- Devices should be able to send their traffic across any number of trunk links to reach their VLAN interface

A VLAN interface will remain in an up/up state so long as there is at least one active port that belongs to the VLAN, or an active trunk link



Review (cont'd)

At Layer 2, you can set various Port configurations including speed, duplex, and mode (access port or trunk port)

When you have both a phone and a PC sharing the same switchport, it is very typical for the phone to plug into the switchport, and the PC to plug into the phone

If the switch supports it, you can then configure the port to belong to both the PC (data) VLAN and the voice VLAN



Review (cont'd)

Switched Port Analyzer (SPAN) is a feature in which you can configure certain ports on a switch to copy all traffic to another port

You typically then connect a sniffer or protocol analyzer to the mirrored port to monitor traffic

A jumbo frame is one in which the Ethernet frame size far exceeds the maximum transmission unit size of 1500 bytes

Most jumbo frames are around 9000 bytes in size

We use jumbo frames on links that require high performance with fewer actual packets being sent

- Such as between a server and a storage controller



Review (cont'd)

When you directly connect two devices together, you must choose between using a straight through cable or a crossover cable

A straight through cable is one in which both ends of the cable have the exact same pinout

A crossover cable is one in which the transmit (orange pair) and receive (green pair) are switched at one end

Because the transmit pins of one interface on one device must connect to the receive pins of the interface on the other device, you must use a crossover cable between devices that have the same pinout

- Router to router, PC to PC, switch to switch, etc.

Traditionally, we only used straight through cables when we connected any other type of device to a switch

- The switchport itself flipped the transmit and receive pairs
- Most modern switches can automatically detect if the wrong type of cable is used, and flip the transmit and receive pairs accordingly



Review (cont'd)

Flow control is a predictive congestion management mechanism in which a switch predicts if it will soon run out of receive buffer space

- In that case, it will send a pause request to the sending device to slow down so it can clear its buffer out

Flow control has largely been replaced by Quality of Service

- Traffic gets a certain priority level depending upon its priority marking

IGMP snooping helps a switch distinguish between different multicast sessions

- It tracks which ports have clients that subscribe to a certain multicast group

IGMP snooping causes a frame to forward multicasting out specific ports, rather than flooding the frame out all ports



Review (cont'd)

Power over Ethernet (PoE) allows the switch to provide DC power to the end device

- Access point, IP phone, camera, industrial controls, alarms, etc.

If your switch does not support PoE, you can either:

- Insert a separate power injector between the switchport and the device
- Or possibly install a power outlet near the device

When configuring PoE, you must be mindful of the total watts of power the switch is capable of delivering

You can configure different priority levels and maximum power delivered to specific ports



Review (cont'd)

Port security allows you to restrict input to switchport by limiting how many and which MAC addresses are allowed to access that port

You can manually configure the allowed MAC addresses for a port

Or you can allow the port to dynamically learn the first MAC address it “hears” as the legitimate MAC address (sticky MAC)

If there is a port security violation, there are three different responses that the port can initiate:

Shut Down is the default port security violation response

- Any violation puts the port immediately into an error- disabled State, effectively shutting it down

Protect does not generate any sort of security notification, nor does it disable the port

- What it does is discard any frames that have unauthorized source MAC addresses

Restrict discards frames with unauthorized source MAC addresses

- Also increments the security violation counter and sends a log message or SNMP trap
- It does not disable the port



Review (cont'd)

MAC aging is the process by which MAC addresses that have become stale after a period of inactivity are automatically purged from the MAC table

You can configure MAC aging globally for all VLANs, or for a specific VLAN

If you configure port security on a switchport that has a voice VLAN, make sure that you set the switch port maximum allowed MAC addresses to 2



Review (cont'd)

Some switch models allow you to toggle a switchport from Layer 2 Layer 3

Switchports that have been changed to Layer 3 are known as Routed Ports

A Routed Port is treated as a standalone Ethernet router interface

- It can receive an IP address and subnet mask, but cannot belong to any VLAN
- It also cannot act as the default gateway for a VLAN

A Routed Port CAN act as the physical default gateway for a subnet

- The switch or hub that the nodes are connected to must itself connect to the Routed Port using a simple uplink
- The routed port has no concept of what a VLAN is



Review (cont'd)

EtherChannel is also known as Link Aggregation or Channel group

EtherChannel is Cisco's implementation of a Link Aggregation Group (LAG)

It allows switches with multiple links to bundle the links together

You can create the EtherChannel either manually or automatically using either PAgp or LACP

PAgp is the Cisco proprietary protocol that attempts to automatically negotiate links into a LAG

LACP is the IEEE standard protocol that attempts to automatically negotiate links into a LAG

Once you have created the EtherChannel, you can then configure how it will be used

It can be an access link, a trunk link, or a routed port link

